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Green digital leadership and algorithmic management for sustainable supply chains: A serial mediation model

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ABSTRACT

Supply chains face resilience and sustainability challenges from disruptions and digital advancements, with limited insight into leadership-driven digital solutions. This study investigates how green digital transformational leadership (G-DTL) influences green digital supply chain transformation (G-DSCT) and resilience (G-DSCR). We examine the serial mediation roles of algorithmic management (ALGM) and green digital absorptive capacity (G-DAC) in these relationships. Utilizing transformational leadership and absorptive capacity theories within the stimulus–organism–response framework, we collected and analyzed survey data from 324 supply chain professionals in Saudi Arabia using partial least squares structural equation modeling. The results confirm that G-DTL significantly and positively impacts G-DSCT and G-DSCR, with ALGM and G-DAC sequentially mediating the G-DTL–G-DSCR relationship. Practically, leaders should proactively invest in developing green digital leadership competencies and adopt ALGM tools to enhance their absorptive capacity. This strategic combination enables organizations to tackle disruptions, ensuring sustainable transformation and supply chain resilience.

1. Introduction

The competitive focus in supply chain management has shifted toward enhancing operational efficiency, as highlighted by Liu et al. (Liu et al., 2023) and Alabdali and Salam (Alabdali & Salam, 2022). The global recognition of green digital supply chain transformation (G-DSCT) has grown, emphasizing their importance for achieving resilience. Stakeholders must prioritize supply chain resilience to obtain mutual benefits, which includes fostering collaboration and employing algorithmic management (ALGM) with key actors such as suppliers, customers, and logistics providers (Richey et al., 2023). This approach improves information exchange (Rejeb et al., 2021), coordinated responses (Nureen et al., 2023), and overall resilience (Belhadi et al., 2024).

Green digital transformational leadership (G-DTL) involves leaders who advocate for and strategically guide the integration of sustainable practices (Alabdali et al., 2024b) and digital technologies (Yu et al., 2021) into supply chains. Such leaders aim to enhance efficiency and profitability while minimizing environmental impact (Y.S. Chen & Chang, 2013). They use green ALGM to leverage data and machine learning to optimize operations, reduce waste, and improve resource management (Parent-Rocheleau et al., 2024). Their leadership ensures digital transformation (DT) aligns with environmental goals, leading to a resilient and sustainable business model (AlNuaimi et al., 2022).

Incorporating ALGM has substantially transformed green and digital supply chains (Lindebaum et al., 2020). This shift has digitalized products, services, and processes, enhancing decision-making and economic performance across various sectors (Ogbeibu et al., 2024). ALGM oversees supply chain activities by analyzing extensive data to improve performance (Neirotti et al., 2021). It is critical for integrating sustainability with efficiency, guiding supply chains to reduce environmental impact while increasing responsiveness and agility (Frederico et al., 2023).

G-DTL is essential as leaders recognize the potential of ALGM to advance green and digital objectives (AlNuaimi et al., 2022). They implement these innovative solutions and ensure that employees align with evolving, technology-driven, environmentally friendly processes (Tosun et al., 2022). The combination of ALGM, transformative leadership, and disruptive green technologies enhances supply chain resilience (Junaid et al., 2023) by building a robust, adaptable, and sustainable network capable of facing challenges while maintaining its commitment to sustainability (Hamdy, 2024).

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G-DSCT involves a comprehensive overhaul of supply chain activities through digital technologies and a focus on sustainability. The transformation aims to improve environmental performance, optimize efficiency, and strengthen resilience against disruptions (Sharma et al., 2024). Leaders dedicated to green DT drive these efforts (AlNuaimi et al., 2022), implementing strategies that merge green principles with digital innovation (Ogbeibu et al., 2024). They promote the adoption of environmentally friendly technologies, such as renewable energy systems, advanced recycling methods, and Internet of Things (IoT) devices for real-time monitoring (Rehman Khan et al., 2022). These technologies are vital for creating an efficient, technologically advanced, sustainable supply chain. Collectively, these elements form a resilient ecosystem capable of adapting to environmental, market, and operational challenges, ensuring long-term sustainability (Ivanov & Dolgui, 2021).

Implementing DT is crucial for modern enterprises. It integrates digital technologies across all organizational aspects to transform operations and enhance customer value. ALGM supports this transformation by making supply chains more agile, transparent, and responsive. According to an Accenture survey (2021), 64 % of supply chain leaders reported an accelerated pace of DT.

Green digital supply chain resilience (G-DSCR) necessitates ongoing adjustments to digital strategies (Yuan et al., 2024). Integrating resilience into DT efforts is crucial, utilizing ALGM and G-DTL (He et al., 2023; Möhlmann et al., 2021). ALGM facilitates supply chains in detecting disruptions, optimizing inventory, and redirecting shipments to mitigate risks (El-Baz & Ruel, 2021). This approach supports robust supplier networks and ensures rapid recovery from disruptions (Boh et al., 2023). Proactive strategies, enhanced by advanced technologies, strengthen resilience against disruptions (Ivanov & Dolgui, 2021). The rapid advancement of ALGM and DT has transformed supply chain management, boosting resilience (Neirotti et al., 2021). Understanding this impact of transformation on supply chain resilience is key to navigating challenges and harnessing digital age benefits (Frederico et al., 2023).

Integrating G-DTL, ALGM, G-DSCT, G-DSCR, and green digital absorptive capacity (G-DAC) highlights interdependencies. Combining ALGM with G-DSCT and G-DSCR is essential for organizational success and adaptability in today's rapidly evolving business environment. Organizations must recognize the importance of ALGM, G-DTL, G-DSCR, G-DAC, and G-DSCT to thrive in the modern business landscape. This capability enables businesses to address disruptions caused by ALGM or other factors, such as G-DAC (Rodríguez-González et al., 2023).

This research is essential due to significant theoretical and practical implications. From a theoretical perspective, research providing an integrated understanding of how leadership-driven DT simultaneously influences sustainability and resilience in supply chains is limited (Esangbedo et al., 2024), leaving scholars without clear conceptual frameworks for understanding these relationships. Practically, managers and organizations face increasing pressure from environmental disruptions and rapid technological advances, making it imperative to understand how strategic leadership and digital capabilities can be leveraged effectively. Without clear insights into these dynamics, organizations risk poor investment decisions, inefficient resource allocation, and failure to achieve long-term sustainability and resilience goals. Therefore, investigating the role of G-DTL and its interplay with ALGM and absorptive capacity is vital for bridging existing theoretical gaps and providing actionable insights for supply chain professionals aiming for sustainable and resilient operations.

To clearly evaluate the success of the proposed model, this research sets specific objectives and metrics. First, we aim to assess the direct and indirect effects of G-DTL on G-DSCT and G-DSCR, using statistically significant path coefficients (β-values) and explanatory power (R^2 values) from structural equation modeling as primary metrics. Second, we evaluate the mediating roles of ALGM and G-DAC by examining the significance of indirect effects and serial mediation paths. Third, we determine the predictive relevance of our model through Stone-Geisser's (Q^2) value, ensuring robust predictive validity. These explicit objectives and metrics provide clear criteria to empirically validate the theoretical relationships and practical utility of the proposed framework.

Despite extensive studies on DT and sustainability in supply chains (Alabdali & Salam, 2022), several critical gaps remain. First, previous research has predominantly focused either on DT or sustainability separately, without providing a clear theoretical integration of both dimensions (Alabdali et al., 2024b; Y.S. Chen & Chang, 2013). Second, while transformational leadership has been well-explored in general organizational contexts, limited empirical attention has been given to its specific role in simultaneously driving both sustainability and digital innovation within supply chain contexts (Alabdali et al., 2024b; Begum et al., 2022; Özgül & Zehir, 2023). Third, understanding of the mediating mechanisms—specifically, the sequential roles of ALGM and absorptive capacity—through which green digital transformation influences both supply chain transformation and resilience is insufficient (Belhadi et al., 2024).

This study explicitly advances existing research by addressing critical limitations identified in previous studies. Specifically, prior research has largely overlooked the integrated role of G-DTL in simultaneously driving sustainability and resilience within supply chains. By clearly examining the sequential mediating roles of ALGM and G-DAC, this research provides a deeper theoretical understanding of the mechanisms enabling supply chain transformation and resilience. Achieving these objectives would clarify previously unclear theoretical relationships, enrich existing conceptual frameworks, and offer practical managerial insights into effectively leveraging leadership and digital capabilities to enhance supply chain sustainability and resilience.

This study examines the intersection of sustainable supply chain practices and digitization, focusing on challenges and opportunities to provide insights on using ALGM and DT to build resilient supply chains. It addresses three key research questions (RQs):

RQ1: How does G-DTL directly impact ALGM, GDAC, GSSCT, and the GDSCR?
RQ2: What is the role of ALGM and GDAC as mediating mechanisms that enable GDTL to enhance GDSCT and G-DSCR?
RQ3: How does G-DSCT channelize the above-mentioned mediated impact of ALGM and G-DAC in causing GDTL to boost G-DSCR?

Moreover, this study addresses a critical gap in the existing literature by investigating how G-DTL facilitates the adoption of ALGM and serves as a strategic enabler in enhancing G-DSCR by integrating sustainable and digital practices. Strengthening organizational capacity to detect, respond to, recover from, and adapt to disruptions involves integrating redundancy, flexibility, collaboration, adaptability, agility, innovation, and advanced technologies (Makhloufi, 2024; Nureen et al., 2023). Organizations must focus on developing robust supply chains while embracing digital innovation to remain competitive and thrive in the dynamic business environment. This approach allows firms to leverage the advantages of ALGM, as demonstrated by Ogbeibu et al. (Ogbeibu et al., 2024). The following subsections will examine each construct individually.

By exploring the interplay of green and digital supply chains with ALGM, DT, and AC, this study aims to clarify enterprises' methods and techniques to fully leverage these advancements. The subsequent sections of this manuscript are organized as follows: the next section presents the theoretical framework, study constructs, and hypotheses; the research methods follow it, the findings, analysis of the findings, and their implications; and the last section concludes this study with limitations and future research directions.

1.1. Theoretical foundation

1.1.1. Stimulus–organism–response (SOR) framework

Introduced by Mehrabian and Russell (Mehrabian & Russell, 1974),

the SOR framework provides a model for understanding how external stimuli influence internal processes and drive organizational behavior. This framework posits that external stimuli (e.g., leadership styles or technological innovations) affect an organization and generate specific responses or outcomes. In this study, the external stimuli are G-DTL and ALGM, which are crucial in guiding how organizations manage environmental challenges and DT (Alabdali et al., 2024b; Dalvi-Esfahani et al., 2023). G-DTL integrates leadership with sustainability and technological innovation, prompting strategic shifts toward greener and more efficient operations. ALGM, however, employs algorithm-driven decision-making to enhance operational efficiency, allowing organizations to adapt swiftly to changing market conditions (Dalvi-Esfahani et al., 2023). Such leadership approaches act as external stimuli that induce internal organizational changes, paving the way for more profound transformations (Alabdali et al., 2024b).

Within the SOR framework, the internal mechanisms—referred to as the organism—include G-DAC and SCT (Qu et al., 2022). G-DAC signifies an organization's capacity to absorb, internalize, and apply new digital and environmental knowledge, vital for integrating sustainable and digital practices. It reflects the organization's readiness to adapt to external pressures, such as regulatory demands or market shifts toward sustainability. Conversely, SCT involves restructuring and optimizing supply chain processes to align with green and digital objectives, ensuring efficient operation while achieving sustainability goals (Akbari & Hopkins, 2022). The ultimate response generated by these internal mechanisms is G-DSCR, which indicates the organization's ability to withstand and recover from disruptions, particularly environmental or technological challenges (Zekhnini et al., 2022). The SOR framework thus provides a comprehensive view of how leadership and management practices, such as G-DTL and ALGM, enhance internal capacities like G-DAC and supply chain operations, leading to improved resilience and sustainability in the supply chain.

1.1.2. Transformational leadership theory

Through their transformational style, leaders initiate transformation and change by evaluating the existing corporate culture and then realigning it toward a new destination. This entails transforming collective values and principles (Bass & Avolio, 1994). This approach is particularly relevant for G-DTL, as it accentuates inspiring workforces to engage in practices that promote environmental sustainability (Begum et al., 2022). The SOR paradigm indicates that the G-DTL style is an external stimulus affecting the internal components of ALGM systems. These internal components (e.g., technological use, digitalization, artificial intelligence (AI), and environmental beliefs) influence organizational resilience (e.g., environmentally responsible behavior in the workplace) (He et al., 2023).

G-DTL is responsible for accelerating green and digital practices within organizations (Esangbedo et al., 2024). Transformational leaders craft a compelling vision integrating sustainability with digital innovation, motivating their organizations to adopt advanced algorithms and data-driven tools. This leadership style aligns organizational goals with technological progress and ensures that resources and support are provided for the effective implementation of ALGM (Majumdar et al., 2024). G-DTL positions the organization to leverage ALGM to optimize decision-making, enhance operational efficiency, and achieve sustainability objectives (Neirotti et al., 2021). Further, such leaders promote cross-functional collaboration, ensuring all departments understand and incorporate green digital technologies into their operations.

Transformational leaders facilitate converting acquired knowledge into actionable strategies and processes. They foster continuous improvement and innovation, enabling the organization to remain agile and responsive to emerging digital and green technologies (Özgül & Zehir, 2023). By cultivating a culture of sustainability and technological advancement, transformational leaders help organizations leverage transformed knowledge to achieve practical outcomes, such as developing new green products, optimizing supply chain processes, and enhancing overall sustainability results (Tosun et al., 2022).

1.1.3. Absorptive capacity (AC) theory

AC refers to an organization's ability to recognize, assimilate, transform, and use external knowledge, particularly at the intersection of green and digital sectors (Cohen & Levinthal, 1990). This concept is crucial for understanding how companies integrate advanced digital technologies and sustainable practices into their supply chains (Tallarico et al., 2024). By leveraging AC, organizations can effectively transform their supply chains sustainably, incorporating ALGM and emerging technologies such as IoT, AI, and big data analytics into current operations, thereby developing actionable strategies (Neirotti et al., 2021). This transformation enhances sustainability and competitive advantage, highlighting the importance of AC in driving G-DSCT and resilience.

AC supports integrating green digital technologies with existing operational frameworks, ensuring smooth adoption across all departments. Effective transformation requires converting assimilated knowledge into practical, actionable strategies that drive continuous improvement and innovation within the supply chain Salam and Bajaba (2023). This involves developing and refining routines that align new insights with existing capabilities. Ultimately, AC enables organizations to achieve tangible outcomes, such as creating new green products (Qu et al., 2022; Bass & Avolio, 1994), optimizing and digitalizing supply chain processes (Salam & Bajaba, 2023), and improving overall sustainability performance. By managing these practices effectively, firms can gain a competitive advantage and contribute significantly to global sustainability goals (Rodríguez-González et al., 2023). This enhanced understanding of AC underscores its critical role in navigating the complex terrain of G-DSCT and G-DSCR, offering a comprehensive framework for strategic decision-making and operational excellence (Junaid et al., 2023; Sarkis et al., 2020).

While existing studies have explored G-DSCT, further analysis is needed on the transformational role of leadership in conjunction with ALGM within the supply chain and its outcomes for establishing a sustainable, digital, and resilient supply chain. This study aims to elucidate the influence of ALGM and DT on the supply chain, highlighting their value and the connections between ALGM and supply chain resilience.

The study intends to make significant contributions by addressing the RQs. It integrates the SOR model (Mehrabian & Russell, 1974) with AC (Cohen & Levinthal, 1990) and TL (Bass & Avolio, 1994) theories to understand the interactions among G-DTL, ALGM, G-DAC, G-DSCT, and G-DSCR. The goal is to provide valuable theoretical and practical insights by linking technology and leadership roles in the green and DT of supply chains, thereby enhancing the understanding of how technological disruptions, revolutions, and sustainability challenges affect green supply chain resilience. The findings will offer insights into resilient supply chain strategies and innovation disruption.

Although prior research has highlighted the importance of leadership in DT, limited attention has been given to its specific role in bridging the adoption of ALGM tools and fostering supply chain resilience. This study extends existing theories by conceptualizing G-DTL as a multidimensional construct that drives the alignment of leadership strategies with algorithmic capabilities, enabling resilient and sustainable supply chain operations in the digital era.

2. Constructs, hypotheses, and their justifications

2.1. Study constructs

The constructs in this study were designed to comprehensively address the intersection of leadership, technological adoption, organizational capacity, and supply chain outcomes in green DT. G-DTL reflects the critical role of leaders in fostering collaboration, driving innovation, and aligning sustainability with digital strategies (AlNuaimi et al., 2022). ALGM captures the integration of data-driven tools and

disruptive technologies to optimize supply chain efficiency and reduce environmental impact (Chatterjee et al., 2023). G-DAC evaluates a firm's ability to acquire, assimilate, and apply external green digital knowledge, which is crucial for leveraging emerging technologies and sustainable practices (Makhloufi, 2024). G-DSCT focuses on adopting green digital practices, such as renewable energy integration and circular economy models, that drive long-term sustainability (Bag et al., 2021). Finally, G-DSCR measures the ability of supply chains to remain robust and adaptive in the face of disruptions while maintaining sustainability goals, an essential consideration considering recent global challenges (Wu et al., 2024). These constructs form an integrated framework addressing critical gaps in green DT and supply chain sustainability literature. The following subsections provide a detailed explanation of each construct.

2.1.1. G-DTL

Integrating eco-friendly and technology-oriented initiatives promotes sustainability through innovative strategies benefiting the environment and society. Digital technology has revolutionized the dissemination of information on environmental issues, improving access to knowledge about sustainability and eco-friendly practices (Y.S. Chen & Chang, 2013). The accessibility of ALGM supports informed decision-making that benefits the environment (Neirotti et al., 2021), enhancing G-DSCT and adopting renewable energy (Lerman et al., 2022). Effective supply chain collaboration necessitates G-DTL, which emphasizes sustainable practices and environmental stewardship (Chatterjee et al., 2023). This leadership style fosters environmentally responsible behaviors among employees, promotes a culture of environmental accountability, and establishes clear sustainability goals (Lyu, 2024).

G-DTL provides essential support and resources for a successful and enduring transition, benefiting organizations and society (Özgül & Zehir, 2023). Despite existing research on G-DTL in various contexts, further investigation is needed into the interplay among ALGM, G-DSCT, G-DAC, and G-DTL and their effects on G-DSCR. Understanding these relationships can offer valuable insights into the strategies, mindsets, and principles necessary for a successful and environmentally conscious digital transition.

2.1.2. ALGM

ALGM employs sophisticated algorithms and technologies, including big data (BD), machine learning (ML), blockchain, augmented reality, AI, and the IoT, to facilitate management decisions with minimal human intervention (Parent-Rocheleau et al., 2024). ALGM involves adopting a digital mindset to support change management and technology integration (Möhlmann et al., 2021). Its autonomous learning and decision-making capabilities enable firms to gain sustained competitive advantages through strategic precision (Lindebaum et al., 2020). ALGM continually analyzes data, learns from errors, and updates its knowledge base to furnish insights (Vision).

ALGM has markedly improved forecasting accuracy, enabling precise projections across various company functions, including supply chain efficiency, production, and labor requirements (Lindebaum et al., 2020). Its application has transformed decision-making processes by leveraging BD to assess service benefits and stakeholder behaviors independent of direct human involvement (Ogbeibu et al., 2024). Integrating ML and AI with ALGM offers reciprocal advantages, enhancing data accessibility and performance, although it also introduces supply chain challenges (Lindebaum et al., 2020; Richey et al., 2023).

2.1.3. G-DAC

Integrating digital and sustainable initiatives is crucial for operational efficiency, innovation, and sustainability in modern supply chains (Zekhnini et al., 2022). G-DAC involves identifying, acquiring, and integrating external knowledge and technology to support these goals (Rodríguez-González et al., 2023). This process includes developing protocols to transform learning into actionable solutions and fostering continuous improvement and innovation (Abou-Foul et al., 2023). Leadership is key to advancing G-DAC by motivating firms to adopt digital and green technologies and aligning knowledge acquisition with strategic objectives (Begum et al., 2022).

ALGM enhances G-DAC by providing data-driven insights and automation, which support information assimilation and process optimization for sustainability goals (Neirotti et al., 2021). Effective management of green digital knowledge allows organizations to adjust their supply chains for greater sustainability and resilience, optimizing processes to incorporate green technologies, improve efficiency, reduce environmental impact, and boost performance (Schilling & Seuring, 2024). G-DAC improves G-DSCR by enabling the swift integration and use of new knowledge to mitigate risks and recover from disruptions (Belhadi et al., 2024; Mishra et al., 2023b).

2.1.4. G-DSCT

A supply chain comprises an interconnected network of organizations, individuals, activities, information, and resources involved in delivering a product or service from supplier to customer (Yuan et al., 2024). SCT involves strategically restructuring processes, systems, and relationships to enhance performance, efficiency, resilience, and adaptability in response to market changes (AlNuaimi et al., 2022). Digital SCT integrates digital and disruptive technologies, such as AI, IoT, and BD analytics, to improve supply chain visibility, efficiency, and decision-making (Nasiri et al., 2020). A green supply chain incorporates environmentally sustainable practices to reduce waste, minimize carbon footprints, and promote resource efficiency (Rahman et al., 2023).

G-DSCT applies digital technology to advance environmental sustainability by fostering eco-friendly innovations, optimizing resource use, and reducing waste (Sharma et al., 2024). This approach involves utilizing smart supply chain technologies, energy-efficient devices, virtual collaboration, and data analytics to enhance resource efficiency (Yuan et al., 2024). G-DSCT is a vital business model transformation that aligns sustainability goals with the supply chain and overall organizational strategies (Schilling & Seuring, 2024). Integrating G-DAC with sustainability aims to achieve ecological goals and necessitates an analysis of its various aspects (Qu et al., 2022). Research can illuminate strategies, challenges, and outcomes of G-DSCT operations, emphasizing leadership, change management, stakeholder engagement, and technological advancements. This study enhances our understanding of G-DSCT's impact on G-DSCR, highlighting the roles of ALGM, G-DTL, and G-DAC in fostering sustainable and technologically advanced supply chains.

2.1.5. G-DSCR

G-DSCR is critical owing to the increasing frequency and severity of global business disruptions (Salam & Bajaba, 2023). It refers to a supply chain's ability to anticipate, prepare for, respond to, and recover from unforeseen events while maintaining connectivity and oversight. G-DSCR addresses vulnerabilities in lean and global supply networks by embedding resilience into supply chain design (Wu et al., 2024). Effective G-DSCR enables organizations to navigate disruptions and restore operations efficiently, emphasizing the importance of risk assessment, collaboration, technological adoption, and continuous improvement. Key characteristics include redundancy, flexibility, collaboration, and information visibility (Salam & Bajaba, 2023).

The significance of G-DSCR is further highlighted by its role in fostering innovative practices and strategic capabilities that enable firms to recover or surpass their pre-disruption state (Shmueli & Koppius, 2011). The demand for G-DSCR drives G-DTL, promoting innovation and resilience in supply chains (Alabdali et al., 2024b). G-DTL generates new opportunities by influencing innovative products, services, or business models despite potential obstacles to supply chain resilience (Y.S. Chen & Chang, 2013). These dynamics affect market disruption, value creation, market expansion, and the evolution of SCT and resilience. The

following subsections discuss the relationships between these constructs.

2.2. Study hypotheses

2.2.1. G-DTL and ALGM

G-DTL is crucial for fostering the development and effective implementation of ALGM within organizations (Ogbeibu et al., 2024). Transformational leaders who emphasize green and digital initiatives establish a vision and culture that drive the adoption of advanced algorithmic tools and data-driven management practices (Alabdali et al., 2024b). By aligning organizational strategies with sustainable DT goals, G-DTL facilitates the seamless integration of ALGM into the organizational framework (AlNuaimi et al., 2022). Such leaders articulate the importance of sustainability and digital innovation, motivating employees to embrace these values and establishing a foundation for ALGM to support environmental and operational objectives (Lindebaum et al., 2020).

Further, G-DTL encourages the proactive pursuit of external knowledge and technological advancements in ALGM, promoting the exploration and adoption of technologies such as AI, ML, and BD analytics (Lyu, 2024). By championing these technologies, transformational leaders ensure that the organization remains at the forefront of digital innovation, equipping it with the necessary tools to enhance efficiency and sustainability in supply chain operations (AlNuaimi et al., 2022). They facilitate the integration of algorithmic tools through cross-functional collaboration and allocate resources for training and development, enabling employees to use ALGM effectively (Ogbeibu et al., 2024). This collaborative approach ensures the widespread adoption of ALGM across the organization, improving overall efficiency and sustainability (Möhlmann et al., 2021; Lindebaum et al., 2020). Transformational leaders convert acquired knowledge into actionable strategies, fostering continuous improvement and leveraging algorithmic insights to optimize supply chain processes, thus maintaining a competitive edge and achieving long-term sustainability goals (Lyu, 2024; Ogbeibu et al., 2024). This hypothesis highlights the pivotal role of leadership in bridging organizational goals with technological advancements. By fostering a culture of innovation and sustainability, transformational leaders enable the strategic adoption of ALGM tools, ensuring environmental and operational objectives are met. Therefore, this study proposes the following hypothesis:

H1. *G-DTL significantly enhances the usage of ALGM.*

2.2.2. G-DTL and G-DSCR

Leadership is key to navigating today's business environment complexities, characterized by rapid technological advancements and growing environmental concerns. G-DTL integrates sustainability and digital innovation into the strategic vision, encouraging organizations to adopt green practices and use digital technologies to achieve sustainability goals (Y.S. Chen & Chang, 2013). By embedding sustainability into core operations and strategic initiatives, G-DTL fosters a culture that values environmental stewardship and technological progress (Alabdali et al., 2024b).

G-DSCR refers to the capacity of supply chains to withstand, adapt to, and recover from disruptions while maintaining sustainability (Sharma et al., 2024). Achieving G-DSCR involves integrating digital technologies to enhance visibility, flexibility, and responsiveness, using AI, IoT, and blockchain for monitoring activities, predicting disruptions, and supporting informed decision-making (Belhadi et al., 2024). The relationship between G-DTL and G-DSCR is crucial for developing robust and sustainable supply chains, as leaders drive initiatives that enhance resilience through real-time insights and predictive capabilities. G-DTL promotes continuous improvement and innovation, aligning leadership strategies with DT to create agile supply chains that advance environmental sustainability and adapt to dynamic challenges (Yu et al., 2021; Hamdy, 2024). This hypothesis emphasizes the importance of leadership in building resilient supply chains that seamlessly integrate sustainability and adaptability. Transformational leaders are central in enabling organizations to address disruptions, enhance operational agility, and achieve long-term environmental objectives. Consequently, this study proposes the following hypothesis:

H2. *G-DTL significantly influences G-DSCR within organizations.*

2.2.3. G-DTL and G-DSCT

Leadership is crucial in driving organizational change, particularly achieving sustainable and efficient operations (Alabdali et al., 2024b). G-DTL emphasizes environmental sustainability and digital innovation by incorporating these elements into strategic practices (Yuan et al., 2024; Sarkis et al., 2020). Leaders who adopt G-DTL cultivate a culture of sustainability, promoting green practices and leveraging digital technologies to meet long-term environmental and operational objectives (Rehman Khan et al., 2022).

G-DSCT focuses on reconfiguring supply chain processes through digital technologies to improve sustainability (Belhadi et al., 2024; Ivanov & Dolgui, 2021). This approach involves implementing AI, IoT, blockchain, and advanced analytics to optimize activities, reduce carbon footprints, and enhance resource efficiency (Yuan et al., 2024). The synergy between G-DTL and G-DSCT is essential for achieving sustainable supply chain goals as leaders promote the adoption of digital innovations aligned with sustainability objectives (Junaid et al., 2023). G-DTL provides strategic direction, resources, and support, fostering collaboration and ensuring DT efforts align with sustainability goals (Rejeb et al., 2021). This hypothesis highlights the strategic role of leadership in transforming supply chains to align with sustainability and digital innovation goals. It ensures that supply chain processes are reconfigured to enhance efficiency, reduce environmental impact, and achieve long-term sustainability objectives. Therefore, the next hypothesis follows:

H3. *G-DTL significantly influences G-DSCT.*

2.2.4. G-DTL and G-DAC

G-DTL is key to developing and using G-DAC within organizations. It promotes a culture of innovation, sustainability, and continuous learning, encouraging the proactive acquisition of external knowledge and the adoption of advanced digital technologies to support environmental goals (Schilling & Seuring, 2024). By fostering a shared vision of sustainability and DT, G-DTL aligns organizational efforts to effectively integrate green and digital initiatives (Tallarico et al., 2024).

G-DTL inspires employees to innovate and embrace new technologies, providing the support infrastructure necessary for acquiring relevant external knowledge in renewable energy, IoT, AI, and BD analytics (Begum et al., 2022). Such leaders facilitate knowledge assimilation by encouraging cross-functional collaboration and the integration of green digital technologies across departments (Makhloufi, 2024). Through effective communication and strategic guidance, transformational leaders help organizations convert assimilated knowledge into actionable strategies, promoting continuous improvement and innovation (Rodríguez-González et al., 2023). This process enables organizations to develop new green products, optimize supply chain processes, and enhance overall sustainability performance (Özgül & Zehir, 2023; Frederico et al., 2023). This hypothesis emphasizes the leadership's role in fostering an organization's ability to acquire, assimilate, and apply knowledge that integrates sustainability and digital innovation. By promoting innovation, collaboration, and strategic alignment, G-DTL strengthens an organization's ability to integrate sustainability with advanced technologies (Alabdali et al., 2024b). Consequently, this study proposes the following hypothesis:

H4. *G-DTL significantly enhances G-DAC.*

2.2.5. ALGM and G-DAC

ALGM employs advanced digital and data-driven technologies to optimize organizational decision-making (Lindebaum et al., 2020). Its integration can significantly enhance G-DAC by providing a framework for acquiring, processing, and utilizing external green digital knowledge (Neirotti et al., 2021; Abou-Foul et al., 2023). ALGM systematically captures information on emerging green technologies and sustainability practices from various sources (Rejeb et al., 2021). Utilizing data analytics, ML, and AI, ALGM facilitates the assimilation of this knowledge into organizational systems, aiding in understanding complex data patterns, trend prediction, and informed decision-making regarding green digital technologies (Bag et al., 2021; Schilling & Seuring, 2024).

ALGM supports the transformation of acquired knowledge into actionable strategies through real-time insights and predictive analytics, promoting continuous process enhancement and innovation in supply chain operations (Ivanov & Dolgui, 2021). It aids in developing new routines that integrate green digital insights with existing capabilities, fostering a culture of ongoing improvement and innovation (Neirotti et al., 2021). By automating decision-making processes, ALGM ensures rapid responses to green digital opportunities and challenges, optimizing supply chain operations and enhancing sustainability performance (Lindebaum et al., 2020). This data-driven approach maximizes sustainability and operational efficiency, offering significant competitive advantages and measurable environmental benefits (Junaid et al., 2023). This hypothesis emphasizes the role of ALGM in strengthening an organization's ability to absorb and implement green digital knowledge. Through data-driven insights, predictive analytics, and automated decision-making, ALGM facilitates the seamless integration of sustainability-focused innovations into organizational processes, driving continuous improvement and environmental performance. Thus, this study proposes the fifth hypothesis as follows:

H5. *ALGM significantly enhances G-DAC.*

2.2.6. ALGM and G-DSCT

ALGM is a transformative force in G-DSCT that leverages advanced algorithms, ML, and BD analytics to optimize supply chain processes per sustainability goals (Sarkis et al., 2020). ALGM enhances the identification and implementation of green digital technologies, improving the ability to collect and analyze extensive data from various supply chain stages (He et al., 2023). This capability is crucial for identifying inefficiencies, forecasting trends, and uncovering opportunities for integrating green technologies. By providing real-time insights and predictive analytics, ALGM supports informed decision-making, aligning operational efficiency with environmental sustainability (Ogbeibu et al., 2024).

Moreover, ALGM effectively integrates sustainable digital technologies into existing supply chain operations through advanced data analytics and automation (Hinkin, 1998). It enables enterprises to model and simulate scenarios to anticipate and address challenges associated with green technologies (Ogbeibu et al., 2024). ALGM fosters continuous improvement and innovation, optimizing supply chain strategies incorporating green digital technologies (Frederico et al., 2023). By automating routine tasks and enhancing decision-making processes, ALGM reduces environmental impact, increases resource efficiency, and improves sustainability performance (Parent-Rocheleau et al., 2024). This hypothesis underscores the pivotal role of ALGM in driving the transformation of supply chains toward greener and more sustainable practices. By leveraging advanced analytics, automation, and predictive modeling, ALGM facilitates the seamless integration of green digital technologies, enabling organizations to optimize processes, enhance resource efficiency, and achieve sustainability objectives. Accordingly, the sixth hypothesis is as follows:

H6. *ALGM significantly enhances G-DSCT.*

2.2.7. ALGM and G-DSCR

ALGM strengthens G-DSCR by reinforcing supply chains against disruptions while ensuring sustainability (Belhadi et al., 2024). It provides a robust framework for anticipating, mitigating, and responding to risks and uncertainties, thus enhancing supply chain resilience (Neirotti et al., 2021). ALGM improves real-time monitoring and analysis of extensive data, which is crucial for identifying potential disruptions and vulnerabilities (Möhlmann et al., 2021). Through predictive analytics, organizations can anticipate environmental risks, supply shortages, or logistical bottlenecks, allowing for proactive measures to maintain operations during adverse conditions (Lindebaum et al., 2020). This foresight and the ability to preempt disruptions are integral to green and digital supply chain resilience and transformation (Rodríguez-González et al., 2023).

ALGM integrates predictive insights into decision-making processes, automating data analysis and providing actionable insights to ensure informed decisions across all organizational levels (Belhadi et al., 2024). This integration enables a coordinated response to disruptions, preserving green digital supply chain practices even during challenges (He et al., 2023). By continuously assimilating new data and adjusting predictions, ALGM ensures agility and adaptability (Schilling & Seuring, 2024). Utilizing real-time data and advanced analytics, ALGM helps organizations refine and enhance supply chain strategies, creating a continuous improvement loop essential for resilience (Ivanov & Dolgui, 2021). Additionally, ALGM automates routine tasks and optimizes responses to disruptions, leading to reduced recovery times, minimized environmental impact, and improved overall supply chain performance, thus reinforcing sustainability commitments (Ogbeibu et al., 2024). Using real-time data, predictive analytics, and automation, ALGM helps organizations identify and address potential disruptions early; it ensures supply chains remain flexible, adaptive, and sustainable, even during unexpected challenges. Thus, the seventh hypothesis is as follows:

H7. *ALGM significantly enhances G-DSCR.*

2.2.8. G-DAC and G-DSCT

G-DAC is vital for G-DSCT, enabling businesses to recognize, integrate, convert, and apply external knowledge of green digital technologies (Rehman Khan et al., 2022; Akbari & Hopkins, 2022). This capability significantly influences an organization's ability to innovate and enhance supply chain sustainability (Sarkis et al., 2020). Firms with high G-DAC actively pursue advancements in renewable energy, sustainable materials, IoT, AI, and BD analytics, thereby maintaining their leadership in technological and sustainability progress. Continuous acquisition and integration of knowledge into organizational processes are essential for aligning these technologies with strategic objectives and operational frameworks (Salam & Bajaba, 2023).

G-DAC transforms acquired information into actionable strategies and processes, fostering the ongoing improvement of supply chain practices (Belhadi et al., 2024; Gölgeci & Kuivalainen, 2020). By integrating green digital insights with existing capabilities, organizations can develop innovative solutions that boost productivity, reduce environmental impact, and enhance ecological sustainability (Rodríguez-González et al., 2023). This interaction promotes a culture of continuous improvement and innovation, enabling firms to build adaptable supply chains responsive to new knowledge and evolving environmental needs (He et al., 2023). Effective utilization of G-DAC results in more sustainable and efficient supply chain operations, providing significant competitive advantages and measurable environmental benefits (Ivanov & Dolgui, 2021). G-DAC helps organizations acquire, integrate, and use knowledge about green digital technologies to improve supply chain practices. By leveraging this knowledge, businesses can innovate, enhance sustainability, and align operations with environmental goals, creating adaptable and efficient supply chains. Consequently, the eighth hypothesis is as follows:

H8. *G-DAC significantly enhances G-DSCT.*

2.2.9. G-DSCT and G-DSCR

In the dynamic business environment, restructuring supply chains through digital technologies is crucial for achieving sustainability and operational resilience (AlNuaimi et al., 2022; Ivanov & Dolgui, 2021). G-DSCT involves reconfiguring supply chain operations using AI, IoT, blockchain, and advanced analytics technologies to improve efficiency, reduce carbon emissions, and optimize resource use (Belhadi et al., 2024). This transformation aims to create agile, efficient, and environmentally sustainable supply chains (Yuan et al., 2024).

G-DSCR refers to a supply chain's capacity to withstand, adapt, and recover from disruptions while maintaining sustainability (Rodríguez-González et al., 2023). Achieving G-DSCR requires the integration of digital technologies to enhance agility, responsiveness, and operational sustainability (Ivanov & Dolgui, 2021). The relationship between G-DSCT and G-DSCR is key to developing resilient and sustainable supply chains, as effective G-DSCT efforts involve advanced digital capabilities that improve visibility, flexibility, and adaptability (Sarkis et al., 2020; Nasiri et al., 2020). AI, IoT, and blockchain enable proactive risk management, better decision-making, and increased resilience (Yuan et al., 2024). This holistic approach ensures supply networks remain adaptable and capable of addressing operational and environmental challenges (Frank et al., 2019; El-Baz et al., 2023). Integrating advanced technologies improves supply chain operations' visibility, flexibility, and adaptability. This transformation enables supply chains to proactively manage risks, recover from disruptions, and maintain sustainability, ensuring resilience in a dynamic business environment. Therefore, the ninth hypothesis is as follows:

H9. *G-DTL significantly influences G-DSCR.*

2.3. Serial mediation relationships

This study investigates four mediation models to explore the complex interrelationships among leadership, technology, AC, and resilience within a sustainable supply chain. The models examine how various mediating variables affect the direct relationship between G-DTL and G-DSCR. Understanding these mediation pathways shows how G-DTL influences G-DSCR and identifies critical factors that enhance supply chain sustainability and resilience.

2.3.1. G-DTL → ALGM → G-DAC→ G-DSCT

G-DTL promotes the integration of ALGM within organizational practices, which enhances G-DAC and drives G-DSCT (Alabdali & Salam, 2022). Transformational leaders promote green and digital initiatives, creating an environment conducive to ALGM adoption by fostering a culture that values sustainability and technological advancement (Junaid et al., 2023; Akbari & Hopkins, 2022). They articulate a strategic vision incorporating advanced algorithms and data-driven tools to achieve environmental and operational goals, thus positioning the organization to adopt ALGM (AlNuaimi et al., 2022).

The implementation of ALGM, driven by TL, enhances an organization's ability to manage and analyze extensive data (Frederico et al., 2023). ALGM facilitates acquiring and assimilating knowledge of external green digital technologies, such as renewable energy, IoT, AI, and BD analytics (Sarkis et al., 2020). By automating data collection and analysis, ALGM strengthens G-DAC, enabling organizations to convert assimilated knowledge into actionable strategies (Tallarico et al., 2024). This process leads to innovative solutions that enhance sustainability and efficiency, such as optimizing resource use and reducing waste. Enhanced G-DAC, supported by ALGM and guided by TL, drives G-DSCT, resulting in more resilient and efficient supply chains aligned with environmental standards and organizational goals (Belhadi et al., 2024). G-DTL fosters a culture of sustainability and innovation, encouraging the adoption of ALGM. ALGM enhances the ability to acquire, process, and apply G-DAC, which drives G-DSCT. This process results in sustainable and efficient supply chains aligned with environmental and organizational goals. Therefore, the serial mediation hypothesis follows:

H10. *G-DTL facilitates ALGM, which enhances G-DAC, ultimately driving G-DSCT.*

2.3.2. G-DTL → ALGM → G-DAC → G-DSCT → G-DSCR

G-DTL initiates a cascading effect that improves ALGM, G-DAC, G-DSCT, and, ultimately, G-DSCR (Alabdali et al., 2024b; He et al., 2023). This process highlights the systematic influence of leadership on sustainable and resilient supply chain practices through advanced digital technologies and knowledge management (Tallarico et al., 2024). G-DTL fosters a culture committed to sustainability and digital innovation, integrating environmental goals with technological advancements (Alabdali & Salam, 2022; Sarkis et al., 2020). Transformational leaders encourage continuous learning and innovation, ensuring effective ALGM adoption (Möhlmann et al., 2021). ALGM optimizes decision-making and enhances G-DAC by incorporating valuable external knowledge into the organization (Ivanov & Dolgui, 2021).

The enhanced G-DAC enables firms to implement new insights into their supply chain operations, resulting in sustainable practices and increased efficiency (Gölgeci & Kuivalainen, 2020). This progression culminates in increased G-DSCR, making supply networks more adaptive and resilient to disruptions (He et al., 2023). G-DTL promotes a culture of sustainability and innovation, enabling the adoption of ALGM. ALGM enhances G-DAC by integrating external knowledge, which supports G-DSCT through sustainable and efficient practices. Ultimately, this transformation strengthens G-DSCR, making supply chains more adaptive, resilient, and environmentally sustainable. Thus, the second serial mediation hypothesis follows:

H11. *G-DTL enhances ALGM, which strengthens G-DAC, driving G-DSCT and, ultimately, leading to greater G-DSCR.*

2.3.3. G-DTL → G-DAC → G-DSCT → G-DSCR

G-DTL enhances G-DAC, driving G-DSCT and yielding G-DSCR. The serial mediation shows visionary leadership's significance in fostering sustainable and resilient supply chains through effective knowledge management and technological integration.

G-DTL begins with leaders advocating for sustainability and digital innovation, motivating teams to adopt green initiatives and advanced technologies (Liu et al., 2023; He et al., 2023). These leaders establish a strategic vision aligned with long-term environmental and operational goals, cultivating a culture of continuous learning and openness to new ideas (Yu et al., 2021). Such leadership enhances G-DAC by systematically acquiring, assimilating, transforming, and exploiting external knowledge related to green digital technologies (Lindebaum et al., 2020). This internal mechanism ensures a comprehensive understanding and integration of new knowledge across departments.

With robust G-DAC, organizations effectively implement G-DSCT, applying insights to develop innovative and sustainable supply chain strategies. This includes optimizing resource utilization, reducing environmental impact, and improving efficiency by integrating green digital technologies into supply chain processes for significant sustainability advancements (He et al., 2023; Schilling & Seuring, 2024). The result, G-DSCR, features supply chains that are more resilient, adaptable to disruptions, and capable of leveraging enhanced knowledge and technology to maintain operational continuity and sustainability (Neirotti et al., 2021; Frederico et al., 2023). G-DTL fosters a culture of sustainability and innovation, motivating teams to embrace green initiatives and advanced technologies. Organizations can acquire, integrate, and apply external knowledge of green digital technologies by enhancing G-DAC. This knowledge drives G-DSCT by enabling innovative and sustainable supply chain strategies, which ultimately strengthen G-DSCR, creating supply chains that are adaptive, resilient, and environmentally sustainable. Therefore, the next serial mediation hypothesis follows:

H12. *G-DTL enhances G-DAC, which drives G-DSCT, ultimately leading to greater G-DSCR.*

2.3.4. G-DTL → ALGM → G-DSCT → G-DSCR

G-DTL initiates a process that leads to the adoption of ALGM, which subsequently drives G-DSCT and culminates in G-DSCR. This underscores the strategic role of leadership in leveraging advanced technologies to foster sustainable and resilient supply chains (Rodríguez-González et al., 2023). Transformational leaders articulate a vision that promotes the adoption of ALGM by emphasizing the integration of sustainability and digital innovation, thus inspiring organizations to embrace advanced algorithms and data-driven tools (Neirotti et al., 2021).

Under G-DTL, adopting ALGM enables systematic data management and optimization of decision-making processes (Esangbedo et al., 2024). ALGM facilitates decisions by analyzing BD, providing real-time insights and predictive analytics essential for transforming supply chain operations (Makhloufi, 2024). This process supports G-DSCT by utilizing data analytics to integrate green digital technologies into supply chain practices, enhancing sustainability and operational efficiency (Rejeb et al., 2021). The result, G-DSCR, features a supply chain that, supported by ALGM, becomes more adaptable and resilient, capable of swiftly responding to disruptions while maintaining sustainability (He et al., 2023). G-DTL inspires the adoption of ALGM by promoting a strategic vision that integrates sustainability and digital innovation. ALGM enables efficient data management and predictive analytics, driving G-DSCT by incorporating green digital technologies into supply chain operations. This transformation strengthens G-DSCR, ensuring supply chains become more adaptable, resilient, and sustainable in the face of disruptions. Thus, the fourth serial mediation hypothesis follows:

H13. *G-DTL facilitates the adoption of ALGM, which drives G-DSCT, ultimately leading to greater G-DSCR.*

Fig. 1 illustrates the interaction between G-DTL, ALGM, G-DAC, G-DSCT, and G-DSCR to conceptualize these hypotheses. It shows how G-DTL affects ALGM and G-DAC, which influence G-DSCT and G-DSCR, enhancing supply chain sustainability and resilience.

To operationalize and ensure empirical rigor, this study employs established frameworks and scales to capture the essence of individual constructs and their associations. G-DTL captures leadership behaviors that inspire and guide employees toward achieving green and DT goals. It emphasizes creating a clear vision, fostering collaboration, and encouraging innovative green and digital initiatives to align individual beliefs with organizational sustainability objectives. Complementing this, ALGM highlights the role of automated systems in optimizing resource allocation, managing schedules, setting performance targets, and informing decision-making processes. These systems ensure supply chain efficiency and/or resilience by dynamically responding to real-time data and linking productivity metrics to operational outcomes. Further, G-DAC focuses on an organization's ability to identify, acquire, and apply external green and digital knowledge. It underscores how mastering and integrating new knowledge drives supply chain adaptability, resilience, and transformation. G-DSCR reflects the ability of supply chains to recover, adapt, and thrive during disruptions. This includes maintaining resource alignment, leveraging disruptions for learning, and preserving operational control and connectivity under adverse conditions. Decisively, G-DSCT highlights adopting digital technologies to enhance supply chain connectivity, sustainability, and collaboration. Core elements encompass the digitalization of processes, data integration across systems, active stakeholder involvement, and collaborative decision-making to achieve long-term supply chain efficiency, resilience, and sustainability. This approach ensures the constructs are theoretically robust and practically applicable, laying the foundation for advancing green DT in supply chains.

3. Research methodology

3.1. Measurement scales

The questionnaire, serving as the measurement instrument, comprised two main sections. The first section contained the scales calibrating the five core constructs using 25 items from established literature (see Appendix A1). The second section collected demographic data, including profession, job level, department, qualifications, organizational size, industry type, country, age, sex, and years of experience (YOE; Appendix Table A2).

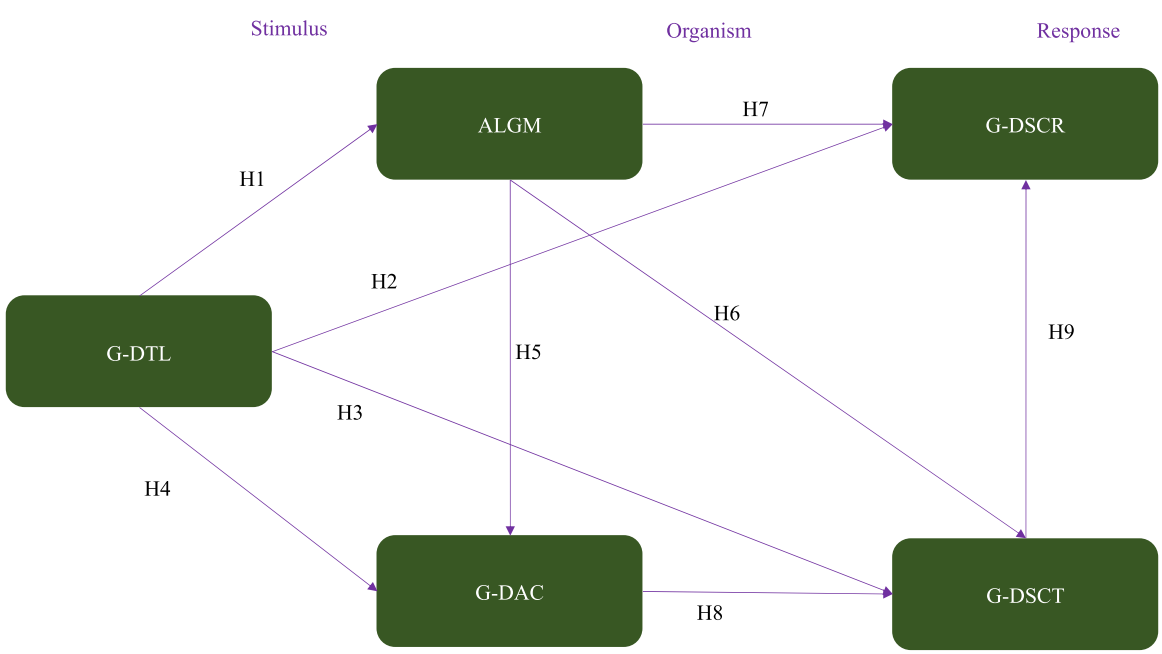

Fig. 1. Conceptual model
Note: G-DTL: green digital transformational leadership; ALGM: algorithmic management; G-DAC: green digital absorptive capacity; G-DSCT: green digital supply chain transformation; G-DSCR: green digital supply chain resilience.

G-DTL was measured using a six-item scale based on Chen and Chang (Y.S. Chen & Chang, 2013; Y.S. Chen & Chang, 2013). ALGM was gauged using five items adapted from Parent-Rocheleau et al. (Parent-Rocheleau et al., 2024), with refinements to align with managing digital tools and algorithms in the sustainability-oriented supply chain context. Similarly, G-DAC was calibrated with four items from Gluch et al. (Gluch et al., 2009), and has been recontextualized to capture the capacity of supply chains to assimilate and apply green digital knowledge and resources. G-DSCT was measured through a six-item scale from Nasiri et al. (Nasiri et al., 2020) and Frank et al. (Frank et al., 2019), focusing specifically on the transformation processes enabled by green digital initiatives. Finally, G-DSCR was measured through a four-item scale from Chunsheng et al. (Chunsheng et al., 2019), El-Baz and Ruel (El-Baz & Ruel, 2021), and El-Baz et al. (El-Baz et al., 2023), emphasizing the role of green digital tools in building resilience within sustainable supply chains. A seven-point Likert scale format, ranging from 1 ("strongly disagree") to 7 ("strongly agree"), was used to record responses, in line with Fornell and Larcker (Fornell & Larcker, 1981).

Recognizing the importance of tailoring measurement scales to the sustainable supply chain context, a two-step validation process was undertaken to ensure the contextual relevance and applicability of the items. First, expert reviews were conducted with a panel of academic and industry professionals with expertise in sustainable supply chain management and DT, following Hardesty and Bearden (Hardesty & Bearden, 2004). They evaluated the items' relevance, clarity, and alignment with this study's context, and their feedback was used to refine the question phrasing and ensure the robustness of the constructs. Second, pilot testing was performed with a subset of supply chain professionals representing the target population, following Hinkin Hinkin (1998) and DeVellis (DeVellis, 2016). This step ensured that the adapted items were understandable and relevant to the participants, with feedback incorporated to improve item clarity and contextual alignment.

The rigorous validation ensures the constructs and items are contextually relevant, theoretically grounded, and empirically robust. Thus, it enables accurate measurement of the sustainable supply chain framework while maintaining alignment with the evolving demands of academia and practice. Before distribution, two industry experts reviewed the questionnaire for relevance and clarity. Additionally, the questionnaire was translated into Arabic and back into English to ensure linguistic accuracy, following Brislin's (Brislin, 1980) cross-cultural research guidelines.

3.1.1. Pilot testing

A pilot test with 40 participants was conducted to ascertain the questionnaire's accuracy and relevance (Reynolds & Diamantopoulos, 1998). The test assessed internal consistency using Cronbach's alpha (CA). Most items exceeded the 0.7 reliability threshold (Cronbach, 1971), confirming their internal consistency, though four items had CA values around 0.554. Items with CA below the conventional threshold of 0.7 can still be retained during pilot testing, particularly in exploratory studies. For example, Nunnally (Nunnally, 1978) and Hair et al. (Hair et al., 2010) suggest that alpha values between 0.5 and 0.6 are acceptable in the early stages, where the primary goal is to refine constructs rather than achieve high reliability (DeVellis, 2016). Moreover, factor loadings greater than 0.5 provide sufficient evidence of an item's contribution to the underlying construct (Field, 2013), making it appropriate to retain such items for further testing.

Given the novelty of the sustainable supply chain context, it is critical to balance methodological rigor with the exploratory nature of this study, allowing the retention of theoretically significant items that may reflect emerging trends or underrepresented dimensions in the literature (DeVellis, 2016). Upon completion of the entire data collection, further analysis was conducted, and it was decided to retain all items, as the CA results met the acceptable reliability threshold, ensuring internal consistency across the constructs. Retaining items based on meeting reliability thresholds is a well-established practice in psychometric research to preserve the theoretical integrity of the constructs and ensure robust measurement (Hair et al., 2010; Taber, 2018).

The pilot test also measured the "coefficient of determination (R^2)," with significant values across constructs: 0.161 for ALGM, 0.725 for G-DAC, 0.582 for G-DSCT, and 0.436 for G-DSCR. These values indicate that the model explained meaningful portions of variance (Henseler et al., 2009), especially for G-DAC and G-DSCT, demonstrating strong predictive power. Participant feedback confirmed the questionnaire's clarity, though minor adjustments were made for improved item comprehension. Feedback from pilot participants was used to refine the questionnaire by addressing issues such as ambiguous wording and improving item clarity in both languages, particularly for complex sustainability constructs. Minor adjustments were made to enhance readability, ensure logical flow, and maintain theoretical integrity, strengthening the instrument for the final data collection. All items were retained for the main study owing to their overall reliability and significant R^2 values. The pilot test established a solid foundation for the final data collection phase as shown in Fig. 2.

3.2. Sampling and data collection

The study specifically targeted supply chain professionals and leaders across various industries in Saudi Arabia to ensure a diverse and representative sample. Saudi Arabia presents a unique context due to its recent phenomenal gains in ensuring green and DT. Its *Vision 2030* initiative emphasizes sustainability, DT, the Saudi Green Initiative (SGI), the global supply chain resilience initiative, and its ongoing efforts to diversify its economy beyond oil dependence (GSCRI 2022; GSCRI 2024; Vision). These factors create a distinct environment where organizations face challenges, such as balancing rapid technological advancements with sustainability goals and opportunities to pioneer green digital practices in supply chain management. The insights from this context provide valuable lessons for similar transitioning economies and developed regions seeking to navigate the complexities of integrating sustainability and DT in (global) supply chains.

The appropriate sample size for the structural model was determined using the *a priori* power analysis method recommended by Soper (Soper, 2023). The minimal sample size was determined to be 193, based on a medium effect size of 0.3 (Cohen, 1992), a statistical power of 0.80, and five latent variables measured by 25 indicators, with a significance level of 0.01. This approach ensured adequate power to detect meaningful effects. The final sample size of 324 respondents exceeded the minimum requirement, enhancing this study's robustness. This larger sample size improved the precision of parameter estimates. It reduced the likelihood of Type I and Type II errors (Gumpili & Das, 2022), contributing to the generalizability and reliability of the findings and reinforcing the validity of the conclusions from the structural model analysis.

A questionnaire-based survey approach was adopted to collect data, with LinkedIn serving as the primary platform for participant recruitment. Convenience and snowball sampling techniques were employed due to practical constraints, including limited access to the targeted professional population of supply chain experts knowledgeable in green digital transformation within Saudi Arabia and the absence of a comprehensive sampling frame for this specific group. Recognized for their efficiency in obtaining timely responses under such circumstances (Alabdali et al., 2024b), these methods were deemed suitable for efficiently identifying respondents with the required experience and expertise. Leveraging LinkedIn, a professional networking platform, for participant recruitment further enhanced the target population's accessibility. This combined approach ensured a relevant, knowledgeable, and accessible sample appropriate for reliably testing the proposed framework within the Saudi Arabian context.

Data collection was conducted via web-based questionnaires, which included a brief explanation of the research objectives to provide participants with an understanding of this study's purpose. One of the

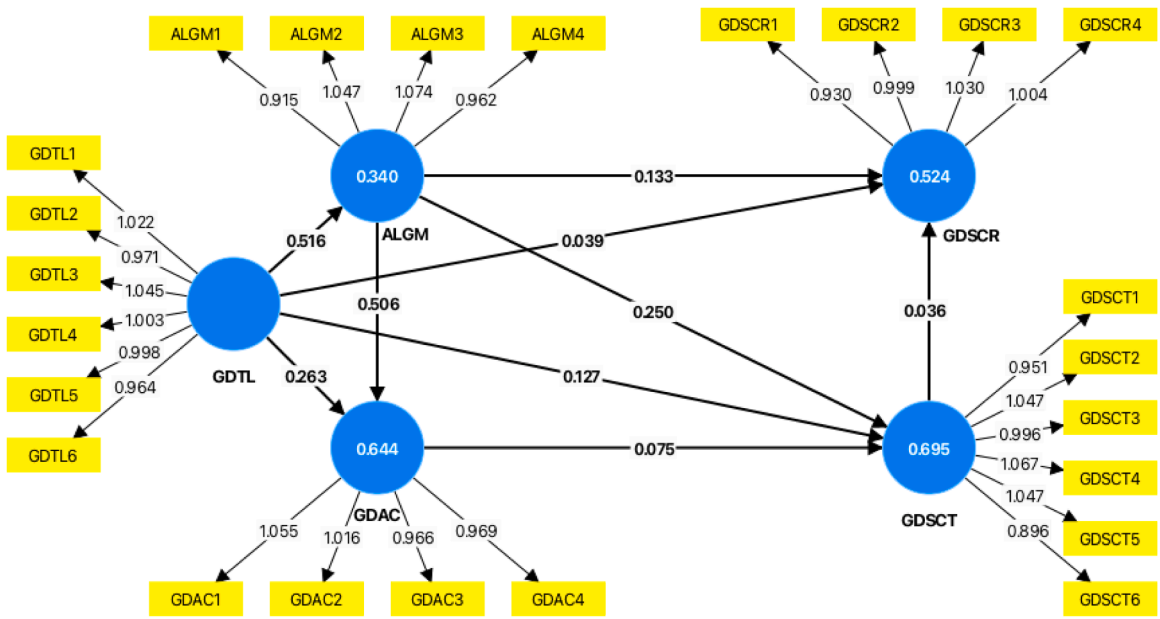

Fig. 2. Pilot testing results.

authors, with over 23 years of professional experience and a LinkedIn network of approximately 30,000 connections, leveraged the platform's messaging feature to distribute personalized invitations to 500 targeted participants. To capture attention and encourage participation, messages were addressed using the recipients' first names while emphasizing the importance of this study.

This approach capitalized on LinkedIn's effectiveness in reaching targeted samples and engaging a broad audience, as supported by prior business research (Alabdali & Salam, 2022). Web-based questionnaires were chosen for their ability to streamline the data collection process, offering a cost-effective, time-efficient method (Sarstedt et al., 2021). To further enhance response rates, follow-up messages were sent as reminders, ensuring higher engagement and completion (Alabdali et al., 2024b).

3.3. Analytical strategy

The study used "PLS-based structural equation modeling" (SEM) to analyze measurement and structural models, employing SmartPLS 4.0 (Ringle et al., 2022). Partial least squares structural equation modeling (PLS-SEM) is particularly suitable when the primary objective of the analysis is to test a theoretical framework from a prediction perspective, as it focuses on maximizing the explained variance of dependent constructs (Hair et al., 2019). Moreover, PLS-SEM is advantageous when dealing with complex structural models that include many constructs, indicators, and relationships, as it efficiently handles such complexity without imposing restrictive assumptions about data distribution (Sarstedt et al., 2021). These characteristics make PLS-SEM an appropriate choice for this study, given its focus on developing and predicting relationships within a sustainability-oriented supply chain framework.

The measured latent marker variable (MLMV) methodology was applied to address concerns related to common method bias (CMB) (Chang et al., 2020). In this approach, YoE was included in the model as an MLMV. YoE was specifically chosen because it is a variable that does not belong to the same domain as the primary constructs being analyzed, ensuring its independence from the theoretical framework (Podsakoff et al.). Prior research has also recommended demographic factors, such as YoE, as effective marker variables owing to their independence from latent constructs being measured (Terho et al., 2022).

The validation of the SEM was conducted by evaluating the measurement and structural models, ensuring the adequacy and robustness of the model in capturing the relationships between constructs (Hair et al., 2019). The measurement model's multicollinearity, reliability, and validity (including convergent and discriminant validity) were assessed using PLS algorithms and bootstrapping techniques, following Hair et al. (Hair et al., 2021). The structural model was then analyzed to evaluate primary, mediating, and serial mediating effects, while PLS blindfolding was used to assess predictive validity.

We followed a three-phase procedure consistent with Anderson and Gerbing (Anderson & Gerbing, 1988). The first phase involved calculating descriptive statistics to summarize the data; the second phase assessed the measurement model's validity and reliability; and the third phase evaluated the structural model's adequacy, ensuring it was appropriate for hypothesis testing and provided reliable insights into the relationships between constructs. This systematic approach ensured a thorough evaluation of both models.

4. Results

4.1. Sample profile

Appendix Table A2 provides a detailed breakdown of the sample. Of the participants, 22 % are women, and 73 % are under 40 years. Half hold senior career positions, with 41 % at the management or senior management level and 11 % at the Head/Director level. Notably, 92 % hold specialist-level positions or higher, and 95 % have attained a bachelor's degree or higher. These demographic details confirm that participants are well-suited for this study, given their experience and professional maturity.

4.2. Common method variance and bias

To address potential "common method variance" (CMV) or "common method bias" (CMB) in the PLS-SEM approach, the researchers employed the "latent marker variable" approach (Chang et al., 2020). Table 1 presents the R^2 values for four models: ALGM, G-DAC, G-DSCT, and G-DSCR. The results indicate that these values remained stable even after incorporating YoE into the analysis. This consistency suggests that the inclusion of YoE did not introduce concerns related to CMB or CMV. Additionally, Fig. 3 visually illustrates the relationship between YoE and the model outputs. The figure demonstrates that the R^2 values remain largely unchanged, further confirming that the inclusion of YoE does not

Table 1
Common method bias and variance.

		Without MV		With MV	
	Q^2	R^2	R^2 adjusted	R^2	R^2 adjusted
ALGM	0.331	0.340	0.338	0.346	0.342
G-DAC	0.460	0.644	0.642	0.647	0.644
G-DSCT	0.464	0.695	0.692	0.696	0.692
G-DSCR	0.321	0.524	0.519	0.537	0.532

Note: MV = marker variable; ALGM = algorithmic management; G-DSCT = green digital supply chain transformation; G-DSCR = green digital supply chain resilience; G-DAC = green digital absorptive capacity.

affect the reliability of the models. These results provide strong evidence that the models maintain their explanatory power without being influenced by potential methodological biases (Terho et al., 2022).

Additionally, the "variance inflation factor" (VIF) was assessed for both inner and outer models, ensuring values remained below 3.3 as shown in Table 2. This validation aligns with Kock's (Kock, 2020) VIF threshold guideline, confirming the absence of CMV issues (Hair et al., 2019).

4.3. Assessment of the measurement model

The measurement model was evaluated by testing multicollinearity, reliability, convergent validity, and discriminant validity, as shown in Table 2 and Fig. 4. The following subsections provide a detailed analysis.

4.3.1. Evaluation of multicollinearity

Collinearity was evaluated by measuring the VIF, with a threshold at VIF < 3.3 (Hair et al., 2019; Hair et al., 2021; Kim, 2019; Shrestha, 2020). Table 2 indicates that all VIF values for the items are below 3.3, indicating the absence of multicollinearity concerns.

4.3.2. Assessment of reliability and convergent validity

Item reliability was evaluated by examining the significance of factor loadings, ensuring they exceeded the recommended threshold of 0.7, indicating strong reliability (Hair et al., 2021). Internal consistency at the construct level was assessed using CA and "composite reliability (CR)." All constructs demonstrated robust internal consistency, with CA and CR values well above the benchmark of 0.7, affirming construct reliability (Hair et al., 2021; Shrestha, 2020). "Convergent validity" was assessed using the "average variance extracted (AVE)." All AVE values exceeded the 0.50 threshold, confirming that the latent constructs explained a substantial amount of variance in the items and indicated adequate convergent validity (Hair et al., 2021; Hair Jr et al., 2020).

4.3.3. Discriminant validity

"Discriminant validity" measures the extent to which a measurement instrument distinguishes between theoretically distinct concepts. This validity was assessed using the Fornell and Larcker (Fornell & Larcker, 1981) method and the "heterotrait-monotrait ratio (HTMT)" approach. As shown in Table 3, discriminant validity is confirmed, with the square root of the AVE consistently exceeding the correlations among latent variables, as per Fornell and Larcker (Fornell & Larcker, 1981). Additionally, all HTMT values were below 0.9, in line with the criteria established by Franke and Sarstedt (Sarstedt et al., 2022) and Rasoolimanesh (Franke & Sarstedt, 2019).

4.4. Structural model analysis

4.4.1. Model fit, explanatory capacity, and predictive precision

Model fit was assessed using the "standardized root mean square residual" (SRMR), with a threshold of <0.08 (Rasoolimanesh, 2022). The SRMR value fell within this range, indicating a satisfactory fit. The "squared Euclidean distance" (dULS) and "geodesic distance" (dG) were within the 95 % and 99 % confidence intervals, respectively, further supporting model fit (Rasoolimanesh, 2022). The "normalized fit index" (NFI), approaching 1, showed a strong match between the hypothesized model and the observed data (Henseler et al., 2015). These metrics—SRMR, dULS, dG, and NFI—collectively show that the model fits the data well and exhibits robust explanatory and predictive capabilities. Table 4 shows the model fit results.

The R^2 indicates the proportion of variance in the endogenous construct explained by external variables (Lohmöller, 2013). According

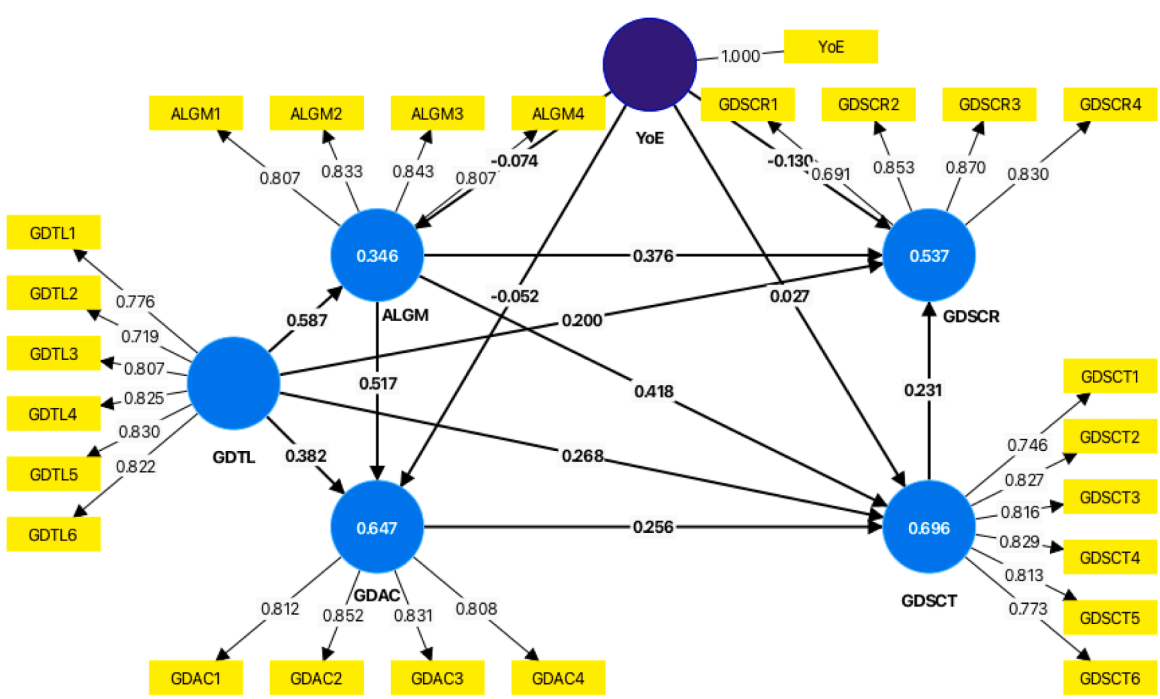

Fig. 3. Marker variable
Note: YoE = participants' years of experience.

Table 2
Reliability and convergent validity.

Construct	Item	FL	VIF	CA	CR	AVE
G-DTL				0.885	0.913	0.636
	G-DTL1	0.776	1.792			
	G-DTL2	0.719	1.594			
	G-DTL3	0.808	2.073			
	G-DTL4	0.825	2.213			
	G-DTL5	0.830	2.283			
	G-DTL6	0.822	2.242			
ALGM				0.841	0.893	0.677
	ALGM1	0.807	1.817			
	ALGM2	0.833	1.907			
	ALGM3	0.842	1.963			
	ALGM4	0.807	1.782			
G-DAC				0.844	0.896	0.682
	G-DAC1	0.812	1.891			
	G-DAC2	0.852	2.128			
	G-DAC3	0.830	1.939			
	G-DAC4	0.808	1.799			
G-DSCT				0.888	0.915	0.642
	G-DSCT1	0.746	1.684			
	G-DSCT2	0.827	2.229			
	G-DSCT3	0.816	2.264			
	G-DSCT4	0.829	2.407			
	G-DSCT5	0.813	2.175			
	G-DSCT6	0.772	1.871			
G-DSCR				0.831	0.886	0.662
	G-DSCR1	0.684	1.540			
	G-DSCR2	0.852	2.178			
	G-DSCR3	0.871	2.134			
	G-DSCR4	0.834	1.808			

Note: FL = factor loadings; VIF = variance inflation factor; CA = Cronbach's alpha; CR = composite reliability; ALGM = algorithmic management; G-DTC = green disruptive technologies; G-DSCT = green digital supply chain transformation; G-DSCR = green digital supply chain resilience; G-DAC = green digital absorptive capacity.

to Henseler et al. (Henseler et al., 2009), R^2 values of 0.75, 0.50, and 0.25 denote strong, moderate, and weak explanatory power, respectively. This study found an R^2 of 0.524 for the endogenous construct G-DSCR, suggesting that approximately 52 % of its variance is explained by external constructs, which is considered adequate given this study's complexity and the inclusion of four predictors (Shmueli & Koppius, 2011).

"Effect size (f^2)" assesses the impact of individual predictors, with thresholds of 0.02, 0.15, and 0.35 for small, medium, and large effects, respectively (Zhang, 2022). In this study, the hypotheses demonstrate a range of effect sizes. Small effects include H2 (G-DTL → G-DSCR) with an f^2 of 0.039, H8 (G-DAC → G-DSCT) with an f^2 of 0.075, and H9 (G-DSCT → G-DSCR) with an f^2 of 0.036. Medium effects include H3 (G-DTL → G-DSCT) with an f^2 of 0.127, H4 (G-DTL → G-DAC) with an f^2 of 0.263, H6 (ALGM → G-DSCT) with an f^2 of 0.250, and H7 (ALGM → G-DSCR) with an f^2 of 0.133. Large effects also include H1 (G-DTL → ALGM) with an f^2 of 0.516 and H5 (ALGM → G-DAC) with an f^2 of 0.506. These results demonstrate the varying degrees of predictor influence within the structural model. Table 5 shows that all values align with Cohen's (Chin, 1998) established thresholds.

"Predictive relevance (Q^2)," as defined by Chin Cohen (2013), evaluates the model's predictive ability. A Q^2 value greater than zero confirms predictive relevance. Using the PLS blindfolding method, the Q^2 values for ALGM, G-DAC, G-DSCT, and G-DSCR were 0.331, 0.460,

Table 3
Discriminant validity.

	ALGM	G-DAC	G-DSCR	G-DSCT	G-DTL
ALGM	**0.823**	*0.878*	*0.787*	*0.881*	*0.675*
G-DAC	0.742	**0.826**	*0.848*	*0.863*	*0.784*
G-DSCR	0.677	0.724	**0.814**	*0.733*	*0.641*
G-DSCT	0.764	0.748	0.657	**0.801**	*0.775*
G-DTL	0.583	0.681	0.574	0.688	**0.798**

Note: ALGM = algorithmic management; G-DTC = green disruptive technologies; G-DSCT = green digital supply chain transformation; G-DSCR = green digital supply chain resilience; G-DAC = green digital absorptive capacity.

Table 4
Assessment of model fitness.

	Saturated model	Estimated model
SRMR	0.061	0.064
d_ULS	1.107	1.225
d_G	0.384	0.407
NFI	0.855	0.849

Note: SRMR = standardized root mean square residual; d_ULS = squared Euclidean distance; d_G = geodesic distance; NFI = normed fit index.

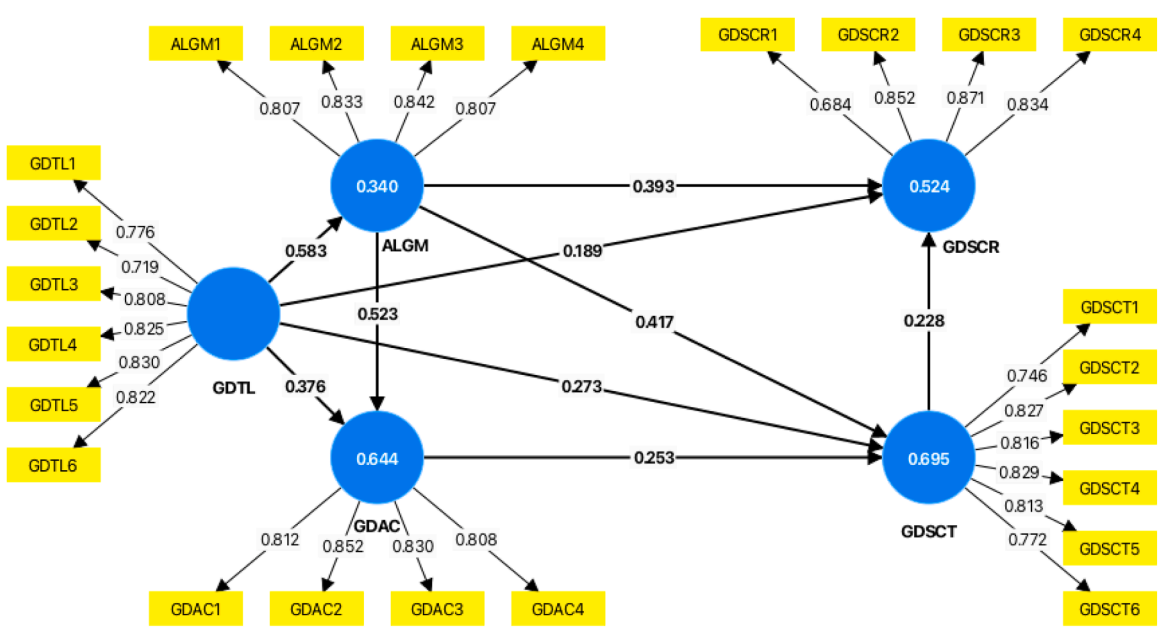

Fig. 4. Measurement model.

Table 5
Assessment of the structural model.

Hypothesis	β	Std Error	t	f^2	CI 2.50 %	CI 97.50 %	p	Decision
H1: G-DTL → ALGM	0.583	0.051	11.473	0.516	0.479	0.677	< 0.001	Supported
H2: G-DTL → G-DSCR	0.189	0.060	3.144	0.039	0.074	0.308	.002	Supported
H3: G-DTL → G-DSCT	0.272	0.054	5.070	0.127	0.168	0.379	< 0.001	Supported
H4: G-DTL → G-DAC	0.376	0.058	6.470	0.263	0.260	0.486	< 0.001	Supported
H5: ALGM → G-DAC	0.523	0.052	10.011	0.506	0.421	0.624	< 0.001	Supported
H6: ALGM → G-DSCT	0.418	0.066	6.368	0.250	0.289	0.549	< 0.001	Supported
H7: ALGM → G-DSCR	0.393	0.064	6.122	0.133	0.269	0.522	< 0.001	Supported
H8: G-DAC → G-DSCT	0.253	0.066	3.832	0.075	0.122	0.380	< 0.001	Supported
H9: G-DSCT → G-DSCR	0.661	0.034	19.226	0.036	0.591	0.726	< 0.001	Supported
H10: G-DTL → ALGM → G-DAC → G-DSCT	0.077	0.022	3.463	-	0.036	0.124	.001	Supported
H11: G-DTL → ALGM → G-DAC → G-DSCT → G-DSCR	0.051	0.016	3.233	-	0.023	0.085	.001	Supported
H12: G-DTL → G-DAC → G-DSCT → G-DSCR	0.063	0.020	3.148	-	0.029	0.106	.002	Supported
H13: G-DTL → ALGM → G-DSCT → G-DSCR	0.161	0.032	4.965	-	0.102	0.229	< 0.001	Supported

Note: ALGM = algorithmic management; G-DTC = green disruptive technologies; G-DSCT = green digital supply chain transformation; G-DSCR = green digital supply chain resilience; G-DAC = green digital absorptive capacity.

0.464, and 0.321, respectively (see Table 1 for the listed Q^2 values). These findings, corroborated by Edeh et al. (2022), Chin (2010), and Wang and Rhemtulla (2021), affirm the model's strong predictive capabalities.

4.4.2. Analysis of the direct and mediating effects

The structural model confirms strong and significant relationships between G-DTL and ALGM, highlighting its critical role in fostering sustainable and resilient supply chain practices. Table 5 presents key statistical indicators, including path coefficients, t-values, and p-values. Fig. 5 illustrates these relationships, showing the strength and significance of each connection. Together, these results validate the impact of G-DTL in driving adaptive and environmentally sustainable supply chain strategies.

G-DTL plays a critical role in driving organizational sustainability and innovation by influencing key components of supply chain management, as reflected in hypotheses H1, H2, H3, and H4. Through its strategic vision and emphasis on sustainability, G-DTL significantly impacts the adoption of ALGM (H1: $\beta = 0.583$, $t = 11.473$, $p < .001$), fostering the integration of advanced technologies and data-driven tools to enhance decision-making and operational efficiency. Further, G-DTL directly contributes to G-DSCR (H2: $\beta = 0.189$, $t = 3.144$, $p = .002$) by promoting adaptive, flexible, and sustainable practices that enable supply chains to withstand disruptions while maintaining continuity. In addition, G-DTL strongly influences G-DSCT (H3: $\beta = 0.272$, $t = 5.070$, $p < .001$) by motivating teams to adopt green digital technologies and optimize resource utilization, leading to improved sustainability and operational efficiency. Moreover, G-DTL enhances G-DAC (H4: $\beta = 0.376$, $t = 6.470$, $p < .001$) by fostering a culture of learning and collaboration, enabling organizations to acquire, assimilate, and apply external knowledge related to green digital innovations.

The analysis identified a strong and significant association between ALGM and G-DAC (H5: $\beta = 0.523$, $t = 10.011$, $p < .001$). This link remained consistent despite additional variables, highlighting ALGM's role in enhancing AC. Moreover, ALGM showed a significant correlation with G-DSCT (H6: $\beta = 0.418$, $t = 6.368$, $p < .001$) and G-DSCR (H7: $\beta = 0.393$, $t = 6.122$, $p < .001$), underscoring ALGM's critical role in SCT and resilience. Additionally, G-DAC exhibited a strong and significant connection with G-DSCT (H8: $\beta = 0.253$, $t = 3.832$, $p < .001$), emphasizing AC's importance in facilitating supply chain change [238]. The association between G-DSCT and G-DSCR was also significant (H9: $\beta = 0.661$, $t = 19.226$, $p < .001$), indicating that transformed supply networks are more resilient.

Noteworthy serial mediation effects were observed. The pathway G-DTL → ALGM → G-DAC → G-DSCT was supported (H10: $\beta = 0.077$, $t = 3.463$, $p = .001$), as was the extended pathway G-DTL → ALGM → G-

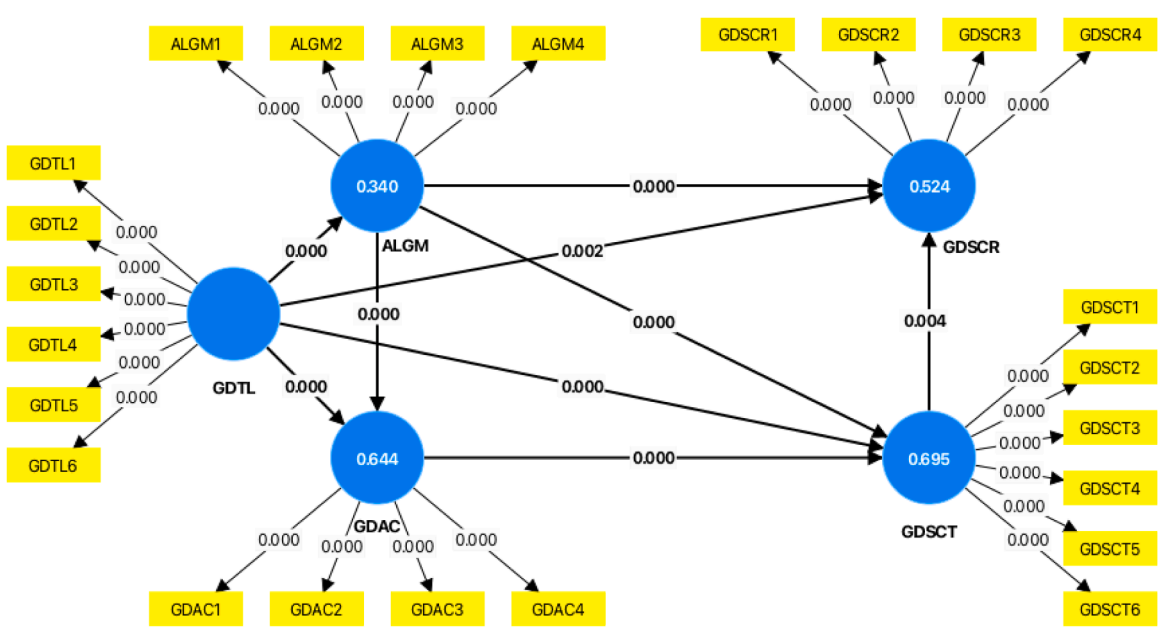

Fig. 5. Structural model.

DAC → G-DSCT → G-DSCR (H11: β = 0.051, t = 3.233, p = .001). Additionally, the mediation effect of G-DAC and G-DSCT in the relationship between G-DTL and G-DSCR was significant (H12: β = 0.063, t = 3.148, p = .002). Hypothesis G-DTL → ALGM → G-DSCT → G-DSCR, was also strongly supported (H13: β = 0.161, t = 4.965, p < .001). Conversely, control factors such as organizational size and age did not correlate significantly with G-DAC or G-DSCT, suggesting minimal impact on these relationships. This analysis highlights the crucial roles of G-DTL, ALGM, and G-DAC in facilitating G-DSCT and G-DSCR (as illustrated in Fig. 6), while indicating that control factors have limited influence.

Table 6 illustrates the impact of incorporating control variables on the relationships among G-DTL, ALGM, G-DAC, G-DSCT, and G-DSCR. Significant findings include a strong positive influence of G-DTL on ALGM (H1: β = 0.583, t = 11.473, p < .001), which remained significant after controlling for other factors. Significant associations were also found between G-DTL and both G-DSCR (H2: β = 0.189, t = 3.144, p = .002) and G-DSCT (H3: β = 0.272, t = 5.070, p < .001), indicating a leader's direct role in enhancing supply chain resilience and change.

5. Discussion and implications

Regarding **RQ1**, the findings confirm that G-DTL positively impacts ALGM, G-DAC, G-DSCT, and G-DSCR, as supported by H1, H2, H3, and H4, respectively. Specifically, H1 demonstrates that G-DTL significantly enhances ALGM, highlighting leadership's role in supporting advanced algorithmic tools and data-driven management, which aligns with Özgül and Zehir (Özgül & Zehir, 2023) and Akbari and Hopkins (Akbari & Hopkins, 2022). H2 confirms the direct positive association between G-DTL and G-DSCR, showing how leadership improves supply chain resilience through strategic alignment and optimal resource allocation (El-Baz et al., 2023). H3 establishes G-DTL's influence on G-DSCT, reflecting its role in driving supply chain transformation by integrating green and digital initiatives. Finally, H4 underscores G-DTL's impact on G-DAC, which demonstrates how leadership enables organizations to integrate green digital knowledge into their practices, fostering a sustainability-oriented culture (Makhloufi, 2024). These findings suggest that organizations should prioritize fostering G-DTL to drive the adoption of green and digital practices, supporting the recommendations of Zekhnini et al. (Zekhnini et al., 2022), Schilling and Seuring (Schilling & Seuring, 2024), and Alabdali et al. (Alabdali et al., 2024b). This will enable efficient supply chain operations, enhanced sustainability, and improved resilience during disruptions (El-Baz & Ruel, 2021). Strategically, organizations can align leadership development programs with sustainability and DT goals to maintain a competitive advantage in volatile markets.

Addressing **RQ2**, the study highlights the mediating roles of ALGM and G-DAC in connecting G-DTL to G-DSCT and G-DSCR. H5 demonstrates the strong positive relationship between ALGM and G-DAC, emphasizing that ALGM enhances an organization's analytics capabilities, enabling it to leverage digital tools for green technologies in line with the findings of Ogbeibu et al. (Ogbeibu et al., 2024). H6 shows ALGM's significant influence on G-DSCT, underscoring how ALGM supports supply chain transformation through timely insights and optimized decision-making (Neirotti et al., 2021). Similarly, H7 establishes ALGM's impact on G-DSCR, highlighting its role in supporting resilience by enabling organizations to effectively recover during disruptions. H8 further confirms that G-DAC facilitates G-DSCT, showing that enhanced absorptive capacity enables organizations to effectively adapt their supply chain processes (Qu et al., 2022; Gölgeci & Kuivalainen, 2020). These findings emphasize the interconnected roles of ALGM and G-DAC in enabling green and digital supply chain transformation and resilience (Abou-Foul et al., 2023). Strategically, organizations should invest in ALGM and analytics capabilities to align with leadership goals, while also building G-DAC to integrate external knowledge. Strategically, focusing on ALGM and G-DAC will enable

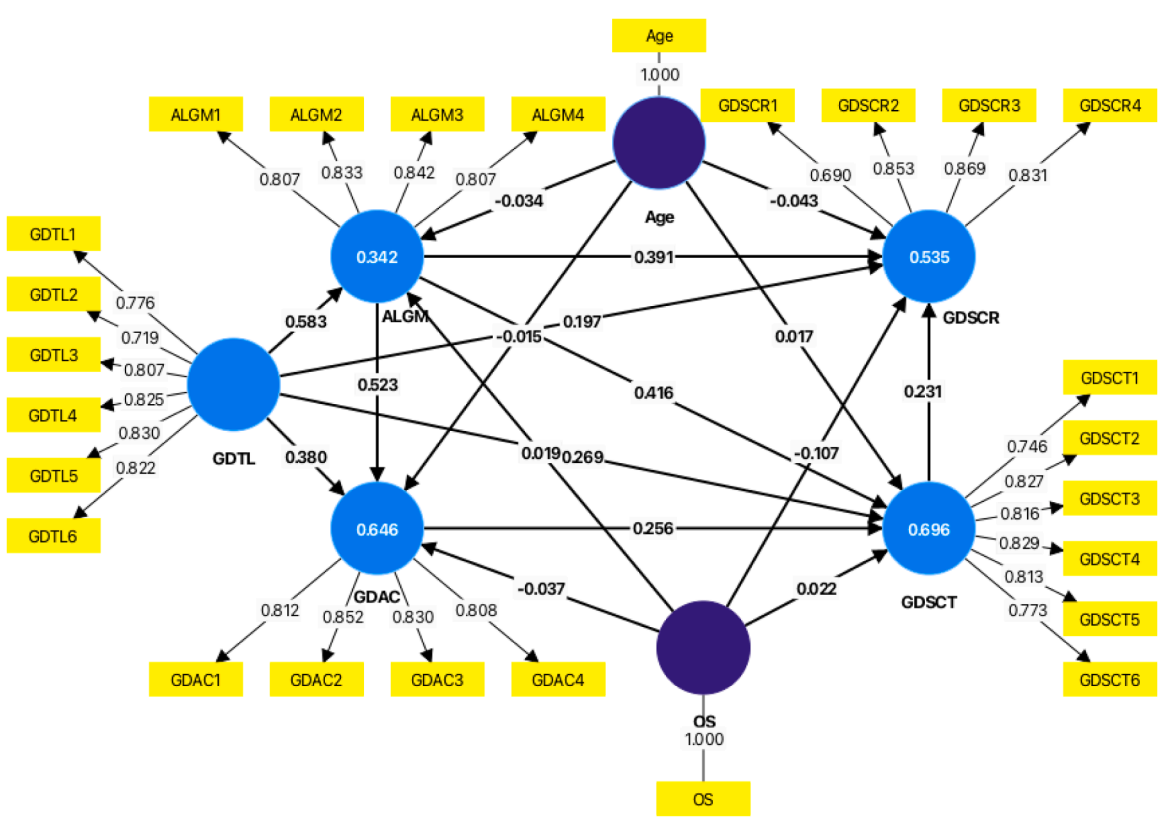

Fig. 6. Control variables
Note: Age = participants' age, OS = organization size.

Table 6
Model comparison: with and without control variables.

	With control variables			Without control variables		
	Original sample	t	p	Original sample	t	p
G-DTL → ALGM	0.583	11.244	< 0.001	0.583	11.473	< 0.001
G-DTL → G-DSCR	0.197	3.304	.001	0.189	3.144	.002
G-DTL → G-DSCT	0.269	4.907	< 0.001	0.272	5.070	< 0.001
G-DTL → G-DAC	0.380	6.499	< 0.001	0.376	6.470	< 0.001
ALGM → G-DAC	0.523	9.949	< 0.001	0.523	10.011	< 0.001
ALGM → G-DSCT	0.416	6.328	< 0.001	0.418	6.368	< 0.001
ALGM → G-DSCR	0.391	6.215	< 0.001	0.393	6.122	< 0.001
G-DAC → G-DSCT	0.256	3.890	< 0.001	0.253	3.832	< 0.001
G-DSCT → G-DSCR	0.231	2.953	.003	0.661	19.226	< 0.001
Age → ALGM	−0.034	0.746	.456			
Age → G-DAC	−0.015	0.432	.666			
Age → G-DSCR	−0.043	1.034	.301			
Age → G-DSCT	0.017	0.561	.575			
OS → ALGM	0.019	0.400	0.689			
OS → G-DAC	−0.037	1.135	0.256			
OS → G-DSCR	−0.107	2.763	0.006			
OS → G-DSCT	0.022	0.635	0.525			

Note: ALGM = algorithmic management; G-DTC = green disruptive technologies; G-DSCT = green digital supply chain transformation; G-DSCR = green digital supply chain resilience; G-DAC = green digital absorptive capacity; Age = participants' age, OS = organization size.

organizations to drive transformation and resilience through data-driven and knowledge-based approaches.

For **RQ3**, the mediating effects observed in H10, H11, H12, and H13 reveal the pathways through which G-DTL indirectly impacts G-DSCT and G-DSCR. H10 demonstrates the pathway G-DTL → ALGM → G-DAC → G-DSCT, showing how leadership-driven G-DAC and ALGM enhance G-DSCT (Makhloufi, 2024; Rodríguez-González et al., 2023). H11 highlights the serial mediation pathway G-DTL → ALGM → G-DAC → G-DSCT → G-DSCR, illustrating how G-DTL indirectly enhances G-DSCR through technology adoption, knowledge integration, and supply chain transformation (Neirotti et al., 2021). Similarly, H12 confirms the pathway G-DTL → G-DAC → G-DSCT → G-DSCR, emphasizing the role of G-DAC and G-DSCT in achieving resilience (Rodríguez-González et al., 2023). Lastly, H13 establishes the pathway G-DTL → ALGM → G-DSCT → G-DSCR, emphasizing the critical role of ALGM in directly enabling G-DSCT and G-DSCR (Neirotti et al., 2021). These pathways underline how G-DTL enhances G-DSCR indirectly through mediating mechanisms (Majumdar et al., 2024; Tosun et al., 2022). These findings suggest that organizations should adopt an integrated approach to leadership, technology, and knowledge management to achieve sustainable and resilient supply chains. Strategically, companies can leverage these pathways to design leadership strategies and DT roadmaps tailored to long-term resilience and sustainability.

Additionally, the stronger direct impacts of G-DTL on ALGM (H1) and G-DAC (H4) compared to its direct impact on G-DSCR (H2) reinforce the importance of these mediators. This finding highlights that G-DTL primarily enhances G-DSCR indirectly by enabling ALGM and G-DAC, which drive G-DSCT and G-DSCR. Furthermore, the strong relationship between G-DSCT and G-DSCR (H9) emphasizes the necessity of continuous transformation to sustain resilience, as G-DSCT serves as a critical channel enabling organizations to maintain operational continuity and sustainability during disruptions (Lerman et al., 2022; Nureen et al., 2023).

Overall, the findings validate the structural model, emphasizing the critical roles of G-DTL, ALGM, and G-DAC in fostering G-DSCT and G-DSCR. Practically, the results underline the importance of G-DTL and advanced digital technologies for transforming supply chains into sustainable and resilient systems. Strategically, organizations can align leadership development, digital innovation, and supply chain strategies to ensure operational success and sustainability in the face of disruptions. This study advances the understanding of sustainable DT by highlighting the interplay between leadership, digital technology, and supply chain performance.

5.1. Theoretical contributions

This study advances the theoretical understanding of green DT, supply chain management, and organizational leadership by exploring the integration of ecological and DT elements within supply chains. It presents a comprehensive framework incorporating key concepts, including G-DTL, ALGM, G-DAC, G-DSCT, and G-DSCR, collectively providing a holistic perspective on how sustainability and digital innovation can be harmoniously applied to achieve superior supply chain outcomes. By synthesizing insights from green transformation and digital innovation, this study develops a novel, multidimensional framework that underscores the strategic importance of leadership in operationalizing advanced technologies and fostering organizational agility, adaptability, and resilience.

Empirical evidence supports the critical role of G-DTL in adopting and implementing advanced digital technologies and sustainable practices. This study highlights the significant impact of G-DTL on ALGM, G-DAC, G-DSCT, and G-DSCR, emphasizing the importance of visionary leadership in guiding organizations toward sustainable and resilient supply chains. This finding contributes to leadership theory by showcasing the distinct capabilities of TL within the green digital era. It demonstrates how G-DTL acts as a catalyst for embedding advanced digital technologies and sustainable practices into the core of supply chain operations. The research further elucidates the importance of ALGM in developing organizational competencies for sustainability and its role in facilitating interactions between G-DTL, G-DAC, and G-DSCT. This interaction enhances the understanding of how algorithmic tools and data-driven decision-making improve the acquisition and use of green digital information, underscoring ALGM's strategic importance in achieving sustainability goals.

Additionally, this study highlights G-DAC's role in enabling SCT, emphasizing robust absorptive capacities to use external green digital knowledge effectively. G-DAC mediates the relationship between ALGM and G-DSCT, thus contributing to the theoretical discussion on AC, especially in green and digital components. The serial mediation analysis reveals that G-DTL influences G-DSCR through intermediary processes involving ALGM, G-DAC, and G-DSCT. This approach refines resilience theory by identifying specific pathways and factors—such as sustainable digital leadership, advanced ALGM practices, and absorptive capacities—contributing to resilience in green DT.

Moreover, this study provides actionable insights into the strategic role of leadership in aligning technological capabilities with sustainability goals. It enhances theoretical insights into pathways that promote supply chain resilience by integrating sustainable digital leadership and advanced technologies with organizational competencies. These insights lay a strong foundation for future research and practical applications in the evolving domain of green DT.

5.2. Practical implications

The results provide actionable insights for supply chain management, sustainability, and organizational leadership. Organizations should prioritize developing G-DTL capabilities. Leaders must communicate effectively and achieve a vision integrating sustainability and digital innovation. This involves investing in executive education and leadership programs focused on environmental stewardship and digital fluency to equip leaders with the skills to guide teams toward sustainable and technologically advanced supply chain operations.

The critical role of ALGM in enhancing both AC and SCT emphasizes that enterprises should adopt advanced algorithmic technologies. Organizations should invest in AI, ML, and BD analytics to improve decision-making processes, optimize operations, and leverage real-time insights. Such investments are crucial for achieving higher efficiency and sustainability in supply chains. To fully benefit from green digital advancements, enterprises must enhance their G-DAC, which involves developing continuous learning and knowledge integration systems, including engaging with external experts, forming industry alliances, and supporting cross-functional teams. Enhancing G-DAC will enable firms to effectively incorporate and use external green digital knowledge for sustainable SCT. The relationship between G-DTL and ALGM in G-DSCT underscores the importance of integrating sustainable practices into supply chain operations. Practitioners should prioritize technologies such as the IoT for monitoring resource consumption, blockchain for transparency, and renewable energy solutions to minimize their environmental impact. These technologies enhance sustainability and improve operational efficiency and resilience.

The study also highlights the importance of G-DSCR. Organizations should implement measures to improve adaptability to disruptions by diversifying supply sources, investing in flexible logistics solutions, and developing resilient risk management frameworks. By focusing on resilience, firms can ensure their supply chains' continuous and stable operation amid environmental and market uncertainties. Finally, this study emphasizes a comprehensive approach to green DT projects, as revealed by the identified interconnected paths. Organizations should avoid isolated system implementations and aim for integrated strategies encompassing leadership, technology, and accountability. This holistic approach ensures that all aspects of the supply chain are aligned to achieve sustainability and resilience. The practical implications suggest that companies can significantly enhance their environmental and operational performance by prioritizing strategic leadership, adopting cutting-edge technologies, improving adaptive capacities, applying sustainable practices, and building resilient supply chains.

6. Conclusion

This study comprehensively explores the dynamic interplay among G-DTL, ALGM, G-DAC, G-DSCT, and G-DSCR, offering new insights into the pathways through which organizations can achieve sustainable and resilient supply chains in the digital era. The findings reveal that G-DTL is pivotal for driving the adoption of advanced ALGM, which catalyzes embedding sustainability and digital innovation into organizational practices. By fostering a culture of proactive leadership, organizations can effectively integrate green digital technologies and streamline external green information, enhancing their G-DAC to assimilate and apply sustainable practices, addressing the RQ1 answer.

The results also highlight the transformative potential of ALGM and G-DSCT in improving supply chain sustainability, efficiency, and resilience. G-DTL emerges as a critical enabler, bridging the gap between strategy and implementation by facilitating the seamless integration of digital and green practices within the context of the supply chain. Moreover, the findings emphasize the importance of cultivating G-DAC to support organizations in navigating the complexities of sustainable green and digital practices. This capacity empowers firms to absorb, adapt, and operationalize external knowledge, ultimately driving innovation and long-term sustainability, which is relevant for answering RQ2.

By examining the serial mediation pathways, this research unveils how the combined influence of G-DSCT and ALGM amplifies the impact of green digital leadership on resilience and AC, which associated with RQ3's answers. It underscores the necessity of a synergistic approach that integrates leadership, technology, and knowledge to enhance operational and strategic outcomes. This holistic perspective strengthens an organization's ability to respond to environmental challenges and aligns DT with sustainability goals.

This study advances theoretical understanding by offering a robust framework combining green and DT concepts in supply chain management. It provides valuable insights for businesses seeking to enhance sustainability and resilience through strategic leadership, advanced algorithmic tools, and strong absorptive capacities. Additionally, this study enriches our understanding of implementing environmentally friendly digital initiatives in supply chains, offering strategic guidance for achieving sustainable and resilient supply chain operations in the digital era.

6.1. Limitations and future research recommendations

This study has some limitations. The cross-sectional design limits the ability to establish causal relationships between variables. Longitudinal studies could provide a deeper understanding of causal dynamics over time. Additionally, reliance on self-reported data may introduce bias; incorporating objective performance measures in future research would enhance validity.

The study's focus on specific industries and geographic regions limits its generalizability. Expanding research to include diverse sectors and international contexts would improve its applicability. Although this study explored the effects of G-DTL, ALGM, and G-DAC on SCT and resilience, future research should examine other potential mediating or moderating factors, such as organizational culture, environmental legislation, and technological infrastructure.

While the findings provide valuable insights into the effectiveness of GDTL as an antecedent and ALGM, G-DAC, and G-DSCT as potential sequential mediating mechanisms to enhance organizational or supply chain performance (specifically, G-DSCR), acknowledging certain econometric limitations is crucial. Particularly, the potential endogeneity between G-DTL and aspects of organizational or supply chain performance requires attention, as bi-directional causalities may exist. For instance, better-performing organizations may be more inclined to adopt G-DTL and ALGM practices rather than practices that solely drive performance improvements. This raises the issue of reverse causality, which must be considered. Future studies could strengthen the framework's robustness by incorporating these considerations.

Moreover, this study primarily used quantitative methods, which may overlook the nuanced insights that qualitative methods could provide. Future research could benefit from qualitative approaches, such as case studies or interviews, to better understand the implementation and experiences of G-DTL, ALGM, and G-DAC within organizations. Investigating emerging technologies, such as blockchain and advanced IoT systems, could further illuminate their impact on supply chain sustainability and resilience. Such studies would offer precise recommendations for integrating cutting-edge solutions into green digital supply chain strategies. Addressing these limitations and employing diverse analytical techniques will enhance our understanding of green DT in supply chains and provide comprehensive strategies for achieving sustainability and resilience.

Data availability

Data will be made available on request.

CRediT authorship contribution statement

Mahmoud Abdulhadi Alabdali: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Muhammad Zafar Yaqub:** Writing – review & editing, Validation, Supervision, Investigation. **Josef Windsperger:** Writing – review & editing, Visualization, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Table A1
Measurement scale.

Construct	Item	Source(s)
Green digital transformational leadership (G-DTL)	G-DTL1. The leader provides a clear environmental and digital technology vision for the followers to follow. G-DTL2. The leader inspires followers with environmental plans to have impact, utilizing digital technologies and sustainable results. G-DTL3. The leader gets the employees to work together for the same green environmental, digital, and sustainable goals. G-DTL4. The leader encourages employees to achieve environmental and digital transformation goals. G-DTL5. The leader acts by considering the environmental beliefs related to green and digital practices of the individuals. G-DTL6. The leader stimulates subordinates to think about green and digital ideas and initiatives.	Chen and Chang (Ogbeibu et al., 2024), Chen and Chang (Y.S. Chen & Chang, 2013).
Algorithmic management (ALGM)	ALGM1. An automated system monitors resource allocation to ensure optimal task completion. ALGM2. Automated systems establish and adjust productivity and efficiency targets based on real-time data. ALGM3. My work schedule is dynamically managed by an electronic system to optimize resource utilization. ALGM4. Performance evaluations incorporate metrics generated by automated systems, emphasizing resource efficiency. ALGM5. Automated systems influence decisions regarding compensation, linked to resource productivity metrics.	Parent-Rocheleau et al. (Parent-Rocheleau et al., 2024)
Green digital absorptive capacity (G-DAC)	G-DAC1. Our organization effectively identifies and assimilates new external green and digital knowledge to enhance supply chain resilience. G-DAC2. We explore and acquire critical green and digital knowledge externally, which is then integrated into our supply chain operations. G-DAC3. The ability to transform and exploit new green and digital knowledge strengthens our supply chain's adaptability and resilience. G-DAC4. We encourage employees to master and apply new green and digital knowledge, driving transformational change within our supply chain.	Gluch et al. (Gluch et al., 2009)
Green digital supply chain resilience (G-DSCR)	G-DSCR1. Our green supply chain is structured to quickly recover and return to its original state after disruptions. G-DSCR2. We maintain a strategic alignment of resources to ensure connectivity among supply chain members during disruptions. G-DSCR3. Our supply chain is designed to retain control over its structure and function under disruptive conditions. G-DSCR4. Our supply chain practices leverage disruptions to extract valuable insights and knowledge, improving future resilience.	Chunsheng et al. (Chunsheng et al., 2019), El Baz and Ruel (El-Baz & Ruel, 2021), El Baz et al. (El-Baz et al., 2023)
Green digital supply chain transformation (G-DSCT)	G-DSCT1. Our supply chain is structured to digitalize processes wherever possible, enhancing efficiency and sustainability. G-DSCT2. We bundle and integrate data from various sources to inform and drive supply chain optimization. G-DSCT3. Our supply chain transformation efforts focus on leveraging digital technologies to strengthen connectivity across business processes. G-DSCT4. We aim to enhance customer engagement by using digital tools to transform the supply chain interface. G-DSCT5. Digital technologies enable seamless information exchange across our supply chain, supporting a collaborative decision-making process. G-DSCT6. We leverage digital solutions to facilitate win-win decisions across all supply chain stakeholders, improving overall feasibility.	Nasiri et al. (Nasiri et al., 2020), Frank et al. (Frank et al., 2019)

Table A2
Analysis of the respondents' profiles and demographics.

Respondents ($N = 324$)	#	%		#	%
Sex					
Male	254	78 %			
Female	70	22 %			
Age (years)			*Supply chain department*		
20–30	72	22 %	Operations	105	32 %
31–40	164	51 %	Logistics	79	24 %
41–50	78	24 %	Procurements	43	13 %
> 51	10	3 %	Warehouse	25	8 %
Experience (years)			Quality	11	3 %
< 1	4	1 %	Planning	42	13 %
2–5	83	26 %	Customer	19	6 %
6–10	76	23 %	*Industry*		
11–15	114	35 %	Food and Beverage	34	10 %
16–20	41	13 %	FMCG	46	14 %
> 21	6	2 %	Retail	34	10 %
Education			Aviation	20	6 %
High school or less	5	1 %	Manufacturing	25	8 %
Diploma	10	3 %	Healthcare	26	8 %
Bachelor's	190	59 %	Telecom	18	6 %
Master's	116	36 %	Others	121	37 %
Ph.D.	3	1 %	*Organization size*		
Occupational level			< 100	15	5 %
Entry level	27	8 %	100–500	156	48 %
Specialist/supervisor	124	38 %	501–1000	46	14 %
Manager/senior manager	132	41 %	1001–5000	57	18 %
Director	35	11 %	> 5001	50	15 %
Leadership	6	2 %			

Note: FMCG = fast moving consumer good.

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